

Solving Classical and Quantum Spin Glasses with Deep Boltzmann Quantum States

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Abstract

We introduce Deep Boltzmann Quantum States, a new quantum extension of deep Boltzmann machines. This architecture extends the physical Hilbert space by introducing layers of hidden quantum spins, hence inheriting efficient block Gibbs sampling and other computational advantages. By combining natural gradients and advanced stochastic optimizers, we successfully obtain the ground state of classical and quantum spin glasses with long-range interactions. When targeting hard ground states of frustrated models, we demonstrate that it is convenient to tune the hardness of the problem Hamiltonian by interpolating from an easy to a hard regime. Hence, we devise Neural Quantum Annealing, a hybrid algorithm that merges the principles of quantum annealing with neural-network wavefunctions to address large-scale optimization. By approximating the adiabatic dynamics on the variational manifold of Deep Boltzmann Quantum States, we obtain exact ground-state solutions of classical and quantum Ising spin glasses with hundreds of spins. Finally, we tackle a real-world NP-complete task known as the Job Shop Scheduling Problem, and solve instances well beyond current limitations of quantum annealing hardware. These findings highlight the effectiveness of integrating quantum annealing principles with the expressive power of modern neural architectures, providing a scalable framework for addressing real-world QUBO problems.

1 Introduction

Optimization is at the core of many scientific, industrial and technological endeavors. They can generally be encoded as *Quadratic Unconstrained Binary Optimization* (QUBO) problems, which may naturally describe problems such as scheduling, resource allocation, and machine learning. Solving large-scale QUBO instances is notoriously difficult, as they often map onto finding the ground-state of complex Ising spin-glass Hamiltonians.

Neural Quantum States (NQS) have emerged as a powerful framework for representing quantum many-body wavefunctions, originally introduced in [1]. By leveraging the high expressivity of Artificial Neural Networks (ANNs), NQS provide flexible variational Ansatz, that combined with the Time Dependent Variational Principle (TDVP), can be used to find the ground-state of a quantum system requiring no prior knowledge about. Such approach has proven effective for approximating ground-state wavefunctions in a variety of quantum spin systems [2].

Quantum Annealing (QA) offers one promising approach to such problems by exploiting the principles of *Adiabatic Quantum Computation* (AQC). In QA, a quantum system is prepared in the easily accessible ground-state of an initial Hamiltonian H_D and then driven out of equilibrium by slowly evolving the initial Hamiltonian into the target Hamiltonian H_T encoding the solution of the problem. However, practical implementations of QA, such as D-Wave quantum annealers [3], face several limitations, including noise, thermal fluctuations, limited qubit connectivity, and scalability constraints, which impede their ability to handle large and complex problems. These limitations, together with the open questions on the role of entanglement and the classical simulability of QA, motivate the study of QA on a restricted variational manifold that can be classically simulated.

Variational Quantum Annealing (VQA) is a variational algorithm originally proposed in [4] to solve ground-state problems of Ising spin glasses.

In this work, we introduce Deep Boltzmann Quantum States (DBQS), a class of variational Ansatz inspired by Deep Boltzmann Machines (BMs). Starting from a physical system composed of N_v quantum spin-1/2 degrees of freedom, which we shall refer to as *visible* spins. We introduce an additional set of N_h *hidden* quantum spin-1/2 variables, interpreted as unobserved degrees of freedom that yield no energy contribution to the system. Then, we use a Jastrow wavefunction to approximate the quantum state of the extended system. As the hidden spins yield no energy contribution, they can be used to model higher-order interactions between the visible spins that are encoded in the entanglement between the visible and hidden spins. Finally, we restrict the interactions in the same way as in deep Boltzmann machines impose limited connectivity. This carefully chosen sparsity keeps the variational problem tractable while still capturing the rich correlation structure that Boltzmann-style hidden layers are known to express.

Then, we introduce *Neural Quantum Annealing* (NQA), a hybrid algorithm to find the ground-state of a target Hamiltonian combining the ideas of VQA and SR with DBQS.

We benchmark our method on instances of the Job Shop Scheduling Problem (JSSP), a representative real-world QUBO problem with high computational complexity. Our results demonstrate the effectiveness of NQA in finding ground-states for Ising spin glasses with sizes up to 700 spins, highlighting its potential for scaling to larger and more complex instances. These findings suggest a promising avenue for addressing real-world optimization challenges by combining neural architectures with advanced quantum-inspired protocols.

We should clearly state what the maximal sizes tackled in the literature with previous quantum Boltzmann machines are. We should highlight that we have significantly improved the addressable sizes, from an Ansatz point of view.

2 Neural Quantum States

Neural Quantum States (NQS) leverage the variational principle, combined with the expressive power of Artificial Neural Networks (ANNs). In this framework, the wavefunction of a quantum many-body system is represented by a parameterized ANN, removing the need for an ad-hoc variational Ansatz based on physical intuition. Originally introduced in Ref. [1], NQS have been successfully applied to solve a variety of quantum ground-states [5, 6, 2, 7] and there are also promising results for the study of quantum dynamics [1, 8, 9, 10, 11]. This section provides the basic background on NQS for spin-1/2 systems necessary for the remainder of this article. For a comprehensive review, see e.g. Ref. [12, 13].

Formally, we consider a system of N spin-1/2 degrees of freedom, whose basis states are labeled by spin configurations $\mathbf{x} = (x_1, x_2, \dots, x_N)$, with $x_i = \pm 1$. This basis is often conventionally associated with the eigenbasis of Pauli-Z operators¹, and the general many-body wavefunction can be expanded as

$$|\psi\rangle = \sum_{\mathbf{x}} \psi(\mathbf{x}) |\mathbf{x}\rangle. \quad (1)$$

In the NQS framework, the wavefunction is expressed as

$$\psi(\mathbf{x}; \boldsymbol{\theta}) \equiv \langle \mathbf{x} | \psi(\boldsymbol{\theta}) \rangle = f_{\text{ANN}}(\mathbf{x}; \boldsymbol{\theta}), \quad (2)$$

where $f_{\text{ANN}}(\mathbf{x}; \boldsymbol{\theta})$ is the output of an ANN that takes $\mathbf{x}$ as an input, and $\boldsymbol{\theta}$ is its set of trainable complex or real parameters.

Leveraging the Time-Dependent Variational Principle (TDVP), it is possible to map the dynamics of the quantum state to a system of ordinary differential equations for the variational parameters [1],

$$i \frac{d}{dt} |\psi(\boldsymbol{\theta})\rangle = H |\psi(\boldsymbol{\theta})\rangle \iff \dot{\boldsymbol{\theta}}(t) = -i S^{-1}(\boldsymbol{\theta}(t)) F(\boldsymbol{\theta}(t)). \quad (3)$$

Here, H denotes the system Hamiltonian, S is the Fisher information matrix (also known as the quantum geometric tensor), and F is the force array, defined as

$$S_{kl}(\boldsymbol{\theta}) = \langle O_k \bar{O}_l \rangle_p - \langle O_k \rangle_p \langle \bar{O}_l \rangle_p, \quad F_k(\boldsymbol{\theta}) = \langle E_{\text{loc}} \bar{O}_k \rangle_p - \langle E_{\text{loc}} \rangle_p \langle \bar{O}_k \rangle_p, \quad (4)$$

with O_k and E_{loc} representing, respectively, the variational derivatives and the local energy

$$O_k(\mathbf{x}; \boldsymbol{\theta}) = \frac{d}{d\theta_k} \log \psi(\mathbf{x}; \boldsymbol{\theta}), \quad E_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) = \frac{\langle \mathbf{x} | H | \psi(\boldsymbol{\theta}) \rangle}{\psi(\mathbf{x}; \boldsymbol{\theta})}. \quad (5)$$

In Eq. (4) we indicated with $\langle \cdot \rangle_p$ the expectation value over the probability distribution

$$p(\mathbf{x}; \boldsymbol{\theta}) = \frac{|\psi(\mathbf{x}; \boldsymbol{\theta})|^2}{\langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta}) \rangle} = \frac{|\psi(\mathbf{x}; \boldsymbol{\theta})|^2}{\sum_{\mathbf{x}'} |\psi(\mathbf{x}'; \boldsymbol{\theta})|^2}. \quad (6)$$

Since the analytical computation of expectation values on a NQS is generally intractable, practical implementations of the TDVP rely on Monte Carlo sampling to obtain empirical estimates of the quantities in Eq. (4). More generally, the expectation value of an operator O can be estimated by computing the empirical average over the Born probability of

$$O_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) = \frac{\langle \mathbf{x} | O | \psi(\boldsymbol{\theta}) \rangle}{\langle \mathbf{x} | \psi(\boldsymbol{\theta}) \rangle}, \quad (7)$$

which is known as the local operator (or local estimator).

Finally, NQS can be employed in the context of ground-state search by carrying out imaginary time evolution. This is formally done by redefining $\tau = it$ in Eq. (3). The intuition behind this approach is that the repeated application of the imaginary time evolution operator $U = e^{-\delta\tau H}$ for short imaginary time steps $\delta\tau$ to an initial state, exponentially suppresses higher-energy components. This results in effectively projecting the state onto the ground-state, provided the initial state has a nonzero overlap with it. This approach is equivalent to minimizing the variational energy

$$E(\boldsymbol{\theta}) = \frac{\langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle}{\langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta}) \rangle} \quad (8)$$

¹We denote the Pauli-Z eigenvalues with $x_i = \pm 1$ for notational consistency with the following sections, since this is identified as the computational basis.

via Natural Gradient Descent [14, 15], an algorithm known as Stochastic Reconfiguration (SR) [16] (see Appendix A).

In practice, the inversion of the Fisher information matrix S is required (as in Eq. (3)), which represents a computational bottleneck. Since S may not be invertible, the Moore-Penrose pseudo-inverse S^+ is often used instead [1, 2, 17], and a small diagonal shift can be applied to the Fisher information matrix to stabilize the numerics. A variant of SR, known as min-SR, has been recently proposed in [2], often offering a significant computational speed-up by requiring the inversion of a smaller matrix. Since these two procedures are formally equivalent, the choice between them is guided only by computational convenience ².

Solving the real-time TDVP dynamics is numerically challenging, primarily due to numerical instabilities arising from multiple sources [13]. A small integration time step δt and a large number of Monte Carlo samples are typically required to accurately solve for real-time dynamics. On the contrary, ground-state search with NQS is defined as the optimization problem in Eq. (8). When solved with SR, the objective is only to approximate the asymptotic state of the imaginary-time TDVP equations rather than to follow the complete imaginary-time trajectory. Hence, to solve the ground-state optimization problem, it is not strictly necessary to use standard SR, and natural gradients can be combined with advanced stochastic optimizers developed for deep learning (see Appendix A), which can dramatically speed up convergence. In this work, we combine natural gradients with momentum [18, 19, 20], as discussed in Section 7.

3 Adiabatic Quantum Computation and practical Quantum Annealing

Quantum Annealing (QA) [21, 22, 23, 24, 25] conceptually relies on the theory of Adiabatic Quantum Computation (AQC) [26]. In this framework, a quantum system is initialized in the easy-to-prepare ground state of a non-interacting *driver* Hamiltonian, often selected as the transverse-field Hamiltonian

$$H_0 = H_x = - \sum_{j=1}^N \hat{\sigma}_j^x . \quad (9)$$

The system is then slowly driven out of equilibrium by tuning an external coupling $s = s(t)$ in an interpolating Hamiltonian

$$H(s) = s H_T + (1 - s) H_0 . \quad (10)$$

The objective is to obtain the ground state of the *target* Hamiltonian H_T , which may either describe a quantum many-body system or encode the solution(s) of a hard classical computational problem in its lowest energy state(s). In an ideal AQC scenario, this is achieved by implementing adiabatic dynamics that, by definition, follow the instantaneous ground-state of $H(s)$, as the coupling $s(t)$ is slowly varied from $s(0) = 0$ to $s(\tau) = 1$ over a sufficiently long total annealing time τ . The system is implicitly assumed to undergo Schrödinger evolution, corresponding to a closed quantum system at zero temperature.

When dealing with classical optimization, the target Hamiltonian is often in the form of an Ising spin glass:

$$H_T = H_z = \sum_{i=1}^N \sum_{j=i+1}^N J_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z + \sum_{i=1}^N h_i \hat{\sigma}_i^z , \quad (11)$$

where the non-zero couplings J_{ij} define an interaction graph among the spin variables and the h_i are local longitudinal fields. This is equivalent to Quadratic Unconstrained Binary Optimization (QUBO) problems and is precisely the type of models that are natively supported on current superconducting QA hardware (e.g., D-Wave systems [3]). Several critical technological and scientific tasks can be reformulated in terms of QUBO [27] or, equivalently, as an Ising spin glass [28], as this represents the prototype of an NP-complete problem.

The theoretical AQC framework encounters severe practical limitations. To avoid exciting higher energy states during the dynamics, a large value of τ is theoretically required by the adiabatic theorem(s) [26]. This value critically depends on the minimum spectral gap Δ encountered during the

²Accordingly, from now on, we will refer to both approaches collectively as SR.

annealing. In simplified terms, the minimum annealing time τ required for adiabaticity typically scales as Δ^{-2} [26]. However, since the annealing path connects a paramagnetic phase in the x direction to a disordered spin glass phase in the z direction, a first-order quantum phase transition is often present in the thermodynamic limit, where the spectral gap closes exponentially fast with the system size N [29]. Hence, the value of τ required for adiabaticity typically diverges exponentially in N , representing a major computational bottleneck for AQC [30, 31, 32].

While quantum annealers closely implement Eqs. (10)–(11), an estimate of the order of magnitude of τ required for adiabaticity is often unfeasible, and practical implementations are also affected by thermal noise, decoherence effects, and hardware constraints [33]. An additional challenge, related to limited hardware connectivity, is the necessity of performing an embedding of the interactions in Eq. (11) into the hardware graph [34]. In practice, current QA hardware does not implement AQC, but is rather employed as a heuristic sampler of the low-energy states of the Hamiltonian in Eq. (11). Drawing a sample requires system initialization, quantum evolution, and (destructive) measurement on the final state, and this whole process is repeated many times (typically thousands of shots). Hence, rather than a single zero-temperature adiabatic evolution, quantum annealers perform several shorter evolutions with a finite value of τ related to device constraints, each of which is often modeled as an open quantum system dynamics [26].

This heuristic approach shows promising results in several applications, especially when utilized as a subroutine in a hybrid quantum-classical algorithm [33, 35] or guided through learning procedures [36], yet no conclusive evidence of quantum speedup [37] to solve a class of models in the category of Eq. (11) has been shown to date. Recent progress [38] has been demonstrated with D-Wave hardware in the regime of fast annealing (τ of the order of a few ns), where coherent Schrödinger dynamics can be realized to simulate the physics of quantum spin glasses on hardware-compatible geometries at the boundary or beyond classical simulations (see also Refs [39, 40] for improved classical simulations). This regime is opposite to the adiabatic theory, and the implications for solving hard *classical* optimization problems as in Eq. (11) remain unclear.

Finally, it is relevant to mention that the abstract framework of AQC can, in principle, be adapted to prepare the quantum ground state of any many-body target Hamiltonian. A straightforward example is represented by a quantum Ising spin glass, simply obtained as in Eqs. (9)–(11), by annealing to a final finite value of transverse field coupling, i.e. to $s(\tau) < 1$. Even more generally, various choices for H_0 and H_T can be envisaged, along with different schedules for the time-dependent couplings. Practical realizations of this broad computational scheme, known as *analog quantum computation*, have been implemented on different experimental platforms, such as trapped Rydberg atom arrays [41].

4 Variational Classical Algorithms Inspired by Quantum Annealing

The concepts and tools of AQC and QA have inspired different classical quantum-inspired variational algorithms that perform real and/or imaginary time evolution on a variational manifold [4, 42, 43]. Moreover, the theoretical framework of AQC has recently been recast into a variational setting [44]. The intuitive idea behind these approaches is that a full Hilbert space evolution may not be necessary to approximate the final ground state. In particular, when the final Hamiltonian is classical and with a glassy energy landscape, as in the typical framework of Eq. (11), obtaining a peaked wavefunction overlapping only with classical lowest-energy configurations may be both unfeasible and unnecessary. In fact, it would be enough to draw a sufficient number of samples from a wavefunction with a finite overlap with one or more of those classical solutions. In this heuristic setting, there is no evidence that a full Hilbert space evolution provides better results than a constrained evolution on a classically simulable variational manifold (see also Fig. 1 in Ref [44]). On the contrary, an evolution on a restricted manifold may even improve results in some cases [42].

In principle, variational simulations of QA can be achieved by solving real-time TDVP Eqs. (3) for a large annealing time τ , as theoretically characterized in Ref. [44] and numerically tested within a Matrix Product State [45] manifold. However, the numerical integration of TDVP equations is computationally demanding for ANNs and would require small time steps, making it extremely challenging for large system sizes and long annealing times τ .

Variational Quantum Annealing (VQA) [4] is a different strategy that relies on iteratively *optimizing*

the variational Ansatz to approximate the instantaneous adiabatic ground-state on the variational manifold. VQA leverages the variational principle to avoid solving real-time TDVP dynamics and may have the potential to mitigate the diverging runtime due to vanishing spectral gaps. Rather than physically evolving a quantum state, one uses a parametrized wavefunction and Variational Monte Carlo to approximately track the ground-state of $H(s)$ in Eq. (10) at a series of $N_A + 1$ discretized points $0 = s_0 < s_1 < \dots < s_{N_A} = 1$ along the annealing schedule. The algorithm realizes a discrete adiabatic path on the variational manifold by iteratively adjusting the Ansatz parameters to minimize the variational energy

$$E(s_j; \boldsymbol{\theta}) = \frac{\langle \Psi(\boldsymbol{\theta}) | H(s_j) | \Psi(\boldsymbol{\theta}) \rangle}{\langle \Psi(\boldsymbol{\theta}) | \Psi(\boldsymbol{\theta}) \rangle}, \quad (12)$$

with $j = 0, \dots, N_A$. Optimizing the Ansatz parameters with SR is equivalent to performing repeated imaginary time evolutions for a set of parametric Hamiltonians $H(s_j)$.

VQA was originally presented in Ref. [4], where the authors devised both a variational version of classical Simulated Annealing and of Quantum Annealing. Both algorithms relied on Recurrent Neural Networks (RNNs) and were tested against standard techniques on several spin glass models. In this setting, VQA with RNNs appeared less effective than its classical counterpart, and VQA results were only shown for polynomial complexity problems, such as the random Ising chain and the 2D Edwards–Anderson model without longitudinal field.

Thanks to its flexibility, the VQA framework can naturally be viewed as an alternative approach to optimizing any NQS for ground-state search. This is useful when a direct optimization with standard SR on the target Hamiltonian H_T is expected to fail due to frustration (as in Eq. (11)), disorder, or other effects. In essence, the hardness of the ground-state problem is tuned from *easy* to *hard* while increasing the parameter s_j , and the NQS is warm-started with the result of the previous iteration (see Appendix A for details).

5 Restricted and Deep Boltzmann machines

Boltzmann machines (BMs) are undirected graphical models originally introduced as a general approach to learning probability distributions over binary variables [46]. BMs are energy-based models over a d -dimensional binary random vector $\mathbf{x} = \{-1, +1\}^d$ with a quadratic energy function ³

$$E(\mathbf{x}; \boldsymbol{\theta}) = -\mathbf{x}^T W \mathbf{x} - \mathbf{b}^T \mathbf{x}, \quad (13)$$

where W and $\mathbf{b}$ are, respectively, the real-valued weight matrix and bias vector that we collectively denote with $\boldsymbol{\theta} = (W, \mathbf{b})$, and the corresponding probability distribution is given by ⁴

$$p(\mathbf{x}; \boldsymbol{\theta}) = Z^{-1} \exp\{\mathbf{x}^T W \mathbf{x} + \mathbf{b}^T \mathbf{x}\}, \quad Z = \sum_{\mathbf{x}'} \exp\{-E(\mathbf{x}'; \boldsymbol{\theta})\}. \quad (14)$$

This class of models becomes particularly useful with the inclusion of hidden (unobserved) variables, which model higher-order interactions between the visible (observed) variables. With the inclusion of hidden variables, BMs become universal approximators of probability functions over discrete variables [47] and have been successfully used to solve a variety of machine learning problems [48, 49, 50]. For a more detailed discussion on standard BMs, we recommend Ref. [46].

Since BMs are notoriously hard to train and their probability distribution is hard to sample, a more practical and widely adopted alternative is the Restricted Boltzmann Machine (RBM), characterized by a single layer of visible variables and a single layer of hidden variables. No intra-layer interactions are included, resulting in a bipartite structure of the interaction graph where the only interactions occur between visible and hidden units. Formally, the visible layer consists of N_v visible variables that we collectively denote with $\mathbf{v}$, and the hidden layer consists of N_h hidden variables that we denote with $\mathbf{h}$. The energy function of the RBM is given by

$$E(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta}) = -\mathbf{v}^T W \mathbf{h} - \mathbf{b}^T \mathbf{v} - \mathbf{c}^T \mathbf{h}, \quad (15)$$

³To be consistent with the standard notation in quantum mechanics, we use spin variables with values ± 1 rather than conventional binary variables. The two conventions are equivalent up to an affine transformation.

⁴To simplify the notation, we omit from now on the explicit dependence on $\boldsymbol{\theta}$ of the partition function Z .

with real-valued trainable parameters $\theta = (W, \mathbf{b}, \mathbf{c})$.

The RBM architecture is generalized by introducing several layers of hidden variables, a model known as Deep Boltzmann Machine (DBM). In contrast to the RBM energy function, the hidden variables in a DBM interact with each other, albeit in a restricted manner, with only adjacent inter-layer connections and no intra-layer connections. By denoting the units in the l -th layer with $\mathbf{x}^{(l)}$, where layer 0 corresponds to the visible units and all hidden layers are indexed for $l > 0$, the energy function of a DBM with N_L layers reads

$$E(\mathbf{x}; \theta) = - \sum_{l=0}^{N_L-2} \mathbf{x}^{(l)T} W^{(l)} \mathbf{x}^{(l+1)} - \sum_{l=0}^{N_L-1} \mathbf{b}^{(l)T} \mathbf{x}^{(l)}. \quad (16)$$

Efficient training of the RBM and DBMs relies on block Gibbs sampling, enabled by the conditional independence of subsets of variables due to the bipartite graph structure. This is discussed in Appendix B.

6 Boltzmann Quantum States

In this work, we introduce a novel framework to extend BMs to quantum variables, designed to represent quantum many-body states. We refer to this formulation as Boltzmann Quantum States (BQS). This class of states possesses a clear physical interpretation, as both hidden and visible variables are promoted to quantum spins over an extended Hilbert space.

Several prior extensions of Boltzmann Machines – particularly RBMs – to quantum settings have been proposed, as briefly reviewed in Appendix C. Our goal is to demonstrate that our approach has two key advantages compared to previous formulations. Firstly, BQS achieve a substantially higher expressive power, thanks to a natural extension from Restricted Boltzmann Quantum States (RBQS) to deep architectures (DBQS). Secondly, these new models naturally inherit block Gibbs sampling, allowing for efficient Monte Carlo estimation even in their deep formulations, with a speedup of orders of magnitude in runtime for the sampling part. In addition to these two key advantages, both RBQS and DBQS exhibit a further distinguishing feature: the calculation of the quantum geometric tensor and any local estimator is greatly simplified. In fact, while the former reduces to the computation of few-point correlators, the latter involves only simple matrix multiplications. This provides an additional computational speed-up for any involved optimization process.

6.1 General formulation

The definition of BQS is based on the idea of introducing unobserved (or hidden) *quantum* spins and using a Jastrow wavefunction to describe the state of the extended system. Physical observables act non trivially only on the Hilbert subspace related to physical (or visible) quantum spins. Formally, given a system of N_v visible spin-1/2 degrees of freedom with a Hamiltonian H_v , a Hilbert space $\mathcal{H}_v$, and computational basis $\{|\mathbf{v}\rangle\}$, we introduce another system of N_h hidden spin-1/2 degrees of freedom, Hilbert space $\mathcal{H}_h$, and computational basis $\{|\mathbf{h}\rangle\}$. We obtain an extended system with Hilbert space $\mathcal{H}_{vh} = \mathcal{H}_v \otimes \mathcal{H}_h$ and computational basis $\{|\mathbf{x}\rangle\} = \{|\mathbf{v}, \mathbf{h}\rangle\}$ on which we consider the Hamiltonian $H_{vh} = H_v \otimes \mathbb{I}_h$. As standard practice in NQS applications, we choose the Pauli-Z basis representation denoted as $|\mathbf{x}\rangle = |x_1, \dots, x_{N_u}\rangle$, with $N_u = N_v + N_h$ and $x_i \in \{-1; +1\}$ corresponding to spin down and spin up, respectively. Finally, a Jastrow variational wavefunction describes the state of the full system, effectively coupling visible and hidden quantum spins:

$$\log \langle \mathbf{x} | \Psi(\theta) \rangle = \sum_{ij} W_{ij} x_i x_j + \sum_i b_i x_i, \quad (17)$$

with $\theta = (W, \mathbf{b})$, where $W \in \mathbb{C}^{N_u \times N_u}$ and $\mathbf{b} \in \mathbb{C}^{N_u}$ are complex variational parameters. We dub this family of Ansatz states as Boltzmann Quantum States (BQS). Note that this wavefunction is not normalized, as commonly done in NQS applications.

It can be easily shown that the probability function defined by a BQS is given by

$$\log[p(\mathbf{x}; \theta)] = \sum_{ij} 2 \operatorname{Re}[W_{ij}] x_i x_j + \sum_i 2 \operatorname{Re}[b_i] x_i, \quad (18)$$

which is the same probability function of a standard Boltzmann machine, with *real* parameters $\theta = (2\text{Re}(W), 2\text{Re}(\mathbf{b}))$. This observation implies that our Ansatz can be interpreted as an over-parameterization of Boltzmann machines when tackling classical problems, namely when the objective is to approximate a target probability distribution. However, when the objective is to approximate the ground-state or the dynamics of a quantum system as outlined in Section 2, the full set of *complex* variational parameters $\theta = (W, \mathbf{b})$ is used to optimize the Ansatz wavefunction in Eq. (17). Hence, BQS are a more general machine compared to standard BMs, as they can capture quantum correlations and entanglement. Compared to other architectures, such as Convolutional Neural Networks, BQS are particularly suitable for models defined on a graph without a clear 1D or 2D geometrical interpretation, or those that lack translational invariance, such as spin glasses or models with disorder.

By definition of the general BQS variational Ansatz in Eq. (17), visible and hidden quantum spins are entangled. As a consequence, it is not generally possible to marginalize the hidden spins to obtain a pure state for the visible ones. Nonetheless, the ground-state expectation value of an observable acting on the physical Hilbert space $\mathcal{H}_v$ can be computed on the ground-state of the extended system $\mathcal{H}_{vh}$ where the BQS is defined. This is discussed in Appendix D, where we also show that restricted and deep BQS architectures reduce to previous proposals of quantum BMs [1, 8] by projecting the hidden units on a specific product state.

6.2 Restricted and Deep Boltzmann Quantum States

Generic BQS suffer from the same intractability as Boltzmann machines. Nonetheless, it is straightforward to implement restrictions on the terms in the Jastrow wavefunction in Eq. (17). This allows to define restricted and deep Boltzmann quantum states, both of which enable efficient block Gibbs sampling and other useful properties.

As in the standard RBM, one can restrict the non-zero coefficients W_{ij} to connect only visible and hidden spins organized in two separate layers, leading to the variational wavefunction for the Restricted Boltzmann Quantum State (RBQS)

$$\langle \mathbf{v}, \mathbf{h} | \Psi(\theta) \rangle = \exp \{ \mathbf{v}^T W \mathbf{h} + \mathbf{b}^T \mathbf{v} + \mathbf{c}^T \mathbf{h} \} . \quad (19)$$

The same notation as in Eq. (15) applies here, where $(\mathbf{v}, \mathbf{h})$ now label the computational basis states for the visible and hidden quantum spins. More generally, in close analogy to the case of DBMs, the spins can be arranged into layers $\mathbf{x}^{(l)}$, with the zero-th layer containing the visible spins, and all the successive layers including hidden quantum spins. By design, non-zero coefficients W_{ij} only connect spins in adjacent layers, and no intra-layer terms are present. Hence, the resulting Ansatz for the Deep Boltzmann Quantum State (DBQS) with N_L layers reads:

$$\langle \mathbf{x} | \Psi(\theta) \rangle = \exp \left\{ \sum_{l=0}^{N_L-2} \mathbf{x}^{(l)T} W^{(l)} \mathbf{x}^{(l+1)} + \sum_{l=0}^{N_L-1} \mathbf{b}^{(l)T} \mathbf{x}^{(l)} \right\} . \quad (20)$$

Note that the DBQS Ansatz includes the RBQS as a special case. This is the class of variational wavefunction utilized in our work.

Due to Eq. (18), the probability distribution for the RBQS and the DBQS is the same as a canonical RBM and DBM, respectively. Hence, it factorizes conditionally on alternate layers, enabling the same efficient block Gibbs sampling routinely used by the machine learning community. A single Markov chain step updates all spins simultaneously, and numerical experiments show that autocorrelation times remain $\mathcal{O}(1)$ as the system size increases, in stark contrast to the superlinear scaling observed with local Metropolis moves. This is shown in Section D.4. Consequently, a speedup of at least an order of magnitude is achieved during both training and inference, allowing us to benchmark our methods for system sizes up to 700 spins.

6.3 Computational advantages

We now outline the other key conceptual innovations and computational advantages enabled by our formulation.

Analytical variational derivatives In a generic BQS formulation, the local derivatives O_k required to estimate the Fisher information matrix (see Eqs. (4)–(5)) reduce to simple products of spin variables and admit an analytical expression

$$\begin{aligned}\frac{\partial}{\partial W_{ij}} \log \langle \mathbf{x} | \Psi(\boldsymbol{\theta}) \rangle &= x_i x_j, \\ \frac{\partial}{\partial B_i} \log \langle \mathbf{x} | \Psi(\boldsymbol{\theta}) \rangle &= x_i.\end{aligned}\tag{21}$$

This represents a striking difference compared to NQS implementations based on feed-forward neural networks, where computing gradients requires backpropagation, introducing additional computational complexity and memory overhead. Our approach avoids backpropagation, offering both conceptual simplicity and improved numerical stability. We note that the empirical Fisher information matrix can be efficiently constructed by computing connected correlation functions through sampling of the BQS wavefunction. For instance, the term of the S matrix associated with the parameters W_{ij} and W_{lm} reads

$$S_{ij,lm} = \langle x_i x_j x_l x_m \rangle - \langle x_i x_j \rangle \langle x_l x_m \rangle,$$
(22)

and similar equations can be easily derived for the other terms.

Evaluation of local operators Evaluating the expectation value of an operator O via Monte Carlo requires computing the local estimator defined in Eq. (7). In the context of spin models, operators can always be decomposed as a linear combination of Pauli strings, and the same holds for their local operators by linearity. For a system composed of N spins, a generic Pauli string operator is written as

$$P = \hat{\sigma}_1^{\alpha_1} \otimes \hat{\sigma}_2^{\alpha_2} \otimes \cdots \otimes \hat{\sigma}_N^{\alpha_N},$$
(23)

where each $\alpha_i \in \{0, x, y, z\}$, and $\hat{\sigma}_i^{\alpha_i}$ denotes either a Pauli operator acting on spin i , or the identity operator $\hat{\sigma}^0 = I$. The local estimator of a Pauli string has the general form of

$$P_{\text{loc}}(\mathbf{x}) = f(\mathbf{x}) \frac{\psi(\tilde{\mathbf{x}}; \boldsymbol{\theta})}{\psi(\mathbf{x}; \boldsymbol{\theta})}$$
(24)

where $f(\mathbf{x})$ is some function of the spin configuration $\mathbf{x}$ and $\tilde{\mathbf{x}}$ is the configuration connected to $\mathbf{x}$ by the Pauli string, i.e. $|\tilde{\mathbf{x}}\rangle \propto P |\mathbf{x}\rangle$.

In NQS applications based on a feed-forward ANN, computing P_{loc} requires one full forward pass for both configurations $\mathbf{x}$ and $\tilde{\mathbf{x}}$. For a BQS, the situation is radically different: inserting the explicit Ansatz of Eq. (17) into the ratio in Eq. (24) yields a simple exponential of differences of quadratic and linear forms

$$\frac{\Psi(\tilde{\mathbf{x}}; \boldsymbol{\theta})}{\Psi(\mathbf{x}; \boldsymbol{\theta})} = \exp \left\{ \sum_{ij} W_{ij} (\tilde{x}_i \tilde{x}_j - x_i x_j) + \sum_i B_i (\tilde{x}_i - x_i) \right\}.$$
(25)

In the specific case of a DBQS, when considering operators that act only on the visible spins, this expression simplifies even further

$$\frac{\Psi(\tilde{\mathbf{x}}; \boldsymbol{\theta})}{\Psi(\mathbf{x}; \boldsymbol{\theta})} = \exp \left[\sum_i (\tilde{x}_i^{(0)} - x_i^{(0)}) \left(b_i^{(0)} + \sum_j W_{ij}^{(0)} x_j^{(1)} \right) \right],$$
(26)

and can be computed with simple tensor operations that can be easily vectorized to further reduce the time complexity of the computation. While the evaluation of local observables involving only the visible spins depends solely on the state of the visible units and the first hidden layer, the full set of hidden units remains necessary for sample generation via Gibbs sampling.

To better illustrate this idea, let us consider the case of the local energy for a spin chain of N sites and the Hamiltonian operator $H = \sum_{i=0}^{N-1} \hat{\sigma}_i^x$. This also represents the only non-diagonal contribution for a standard QA Hamiltonian, as shown below in Eq. (10). In this case, the local energy of a given configuration, as defined in Eq. (5), reads

$$E_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) = \sum_{i=0}^{N-1} \frac{\psi(x_0, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_{N-1}; \boldsymbol{\theta})}{\psi(\mathbf{x}; \boldsymbol{\theta})}.$$
(27)

Hence, in a NQS implementation based on a feed-forward neural network, one needs to compute the output of the network for all possible single spin flips, requiring a total of $N + 1$ evaluations of the network forward function. On the other hand, in the case of a DBQS, the local energy is simply given by

$$E_{\text{loc}}(\mathbf{x}; \boldsymbol{\theta}) = \sum_{i=0}^{N-1} \exp \left[-2x_i^{(0)} \left(b_i^{(0)} + \sum_j W_{ij}^{(0)} x_j^{(1)} \right) \right]. \quad (28)$$

Therefore, all the quantities needed to compute the local energy can be evaluated with a single matrix multiplication.

7 Neural Quantum Annealing

Neural Quantum Annealing (NQA) is a novel algorithm that merges the high-level framework of VQA (see Section 4) with the expressive capabilities and sampling efficiency of DBQS (see Section 6), which are iteratively optimized with natural gradients and advanced stochastic optimizers. An illustrative representation of our algorithm is shown in Fig. 1, where the typical NQA optimization trajectory can be visualized on a reduced-dimensional space for a toy-size classical spin model. A preliminary version of our work using standard RBM wavefunctions was presented in [43]. In the following, we describe our algorithm, highlighting its new features and the underlying theoretical motivations that distinguish our approach. A detailed pseudo-code and additional information on hyper-parameters are reported in Appendix E. Central to our method is the exploitation of the scalability and trainability of DBQS.

DBQS Ansatz The primary challenge in sampling low-energy configurations of spin glasses lies in their highly rugged energy landscape, which results from energy frustration and competing interactions. This leads to a multitude of local minima that can trap heuristic optimization methods. In particular, Markov Chain Monte Carlo (MCMC) algorithms relying on local update rules are effectively non-ergodic in the spin-glass phase and suffer from exponentially slow mixing due to large free-energy barriers [51], calling for more refined schemes [52]. Since most modern NQS architectures rely on Metropolis-based MCMC (local update rules), they are expected to perform poorly on spin glass sampling.

Here, we employ DBQS as a variational Ansatz. Unlike other NQS architectures, DBQS enable efficient block Gibbs sampling, which, thanks to its global update rule, is better suited for handling multimodal distributions in glassy energy landscapes. Moreover, DBQS can encode long-range, non-equivariant correlations on random graphs, providing both the expressivity and the sampling capabilities required for the problem. This makes our algorithm viable for large-scale, complex spin glass problems where sampling is the main computational bottleneck. An alternative approach was proposed in Ref. [4], which employs recurrent neural network (RNN) wavefunctions that allow for autoregressive sampling. Although the RNN yields exact, uncorrelated samples, its capacity to represent the quantum many-body wavefunction during the annealing is uncertain, especially for models on a random graph that lacks any 1D or 2D geometry, as the network needs to learn non-equivariant, long-range correlations. In practice, RNNs were found to be more conveniently optimized by thermal annealing [4]. This stands in contrast to several previous studies that have demonstrated the enhanced effectiveness of *simulated* quantum fluctuations over *simulated* thermal fluctuations [23, 53, 54, 55, 56] in exploring rugged energy landscapes of classical frustrated models.

Persistent parallel Gibbs chains Building on the sampling capability unlocked by the DBQS Ansatz, we run an ensemble of persistent Markov chains throughout the entire variational optimization. While a straightforward approach would be to generate and thermalize a new set of Markov chains at each annealing step $j = 0, \dots, N_A$ in Eq. (12), this would be a prohibitively slow procedure for high-dimensional glassy distributions.

Because successive Hamiltonians $H(s_j)$ and $H(s_{j+1})$ change only slightly, it is expected that thermalized chains at step j remain near equilibrium for the new distribution. Consequently, instead of fully re-thermalizing them, one can resume the chains from their last states and apply a few block Gibbs updates to adapt to the updated wavefunction parameters.

SR augmented with momentum As anticipated in Section 2, natural gradients can be combined with advanced stochastic optimizers, enhancing the standard SR algorithm for ground-state search. Rather than a straightforward optimization of the variational energy in Eq. (12), NQA combines natural gradients with a momentum term to perform the parameter updates. When using momentum [18, 19, 20], a velocity vector stores an exponentially decaying average of past gradients. Let us focus on a single annealing step j : for a fixed value of s_j , we perform a sequence of $n = 1, \dots, N_U$ optimization epochs to iteratively minimize the variational energy in Eq. (12). Concretely, at epoch n the updates are

$$\boldsymbol{\nu}_{n+1} = \mathbf{g}_n + \mu \boldsymbol{\nu}_n, \quad \boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n - \eta \boldsymbol{\nu}_{n+1}, \quad (29)$$

where $\mathbf{g}_n$ is the natural gradient, $\boldsymbol{\nu}$ is the velocity vector, η is the learning rate, and $0 \leq \mu < 1$ is the momentum coefficient. The velocity term damps oscillations in directions where the gradient sign alternates while accelerating progress along directions with a persistent sign, often yielding faster and more stable convergence than standard gradient descent [57]. Crucially, both the optimized variational parameters and the final velocity obtained at annealing step j are carried over to warm start the DBQS Ansatz and the optimizer in the next annealing step $j + 1$.

Relaxed adiabaticity on a manifold While in the ideal limit – infinitely many annealing steps N_A , optimization epochs per annealing step N_U , and a sufficiently expressive Ansatz – VQA would recover the true adiabatic path in the whole Hilbert space, it is clearly necessary to set a finite computational budget and select a NQS manifold with finite expressivity. The VQA framework is designed to remain close to the instantaneous variational ground-state, which is defined as the state that minimizes Eq. (12) as $H(s)$ slowly interpolates between H_0 and H_T . As recently pointed out in [44], it is essential to notice that the instantaneous *variational* ground-state does not need to coincide with the *exact* instantaneous ground-state of $H(s)$ at intermediate points s_j . Hence, the discretized manifold evolution does not, in general, closely approximate the ideal adiabatic dynamics in the full Hilbert space. Nonetheless, if the variational manifold is sufficiently expressive to capture both the initial ground-state of H_0 and the target ground-state of H_T , this manifold evolution may yield a valid solution – either the target quantum ground-state, or a wavefunction with finite overlap with classical energy minima of Eq. (11).

Additionally, even when approximating the adiabatic evolution on the manifold rather than in the whole Hilbert space, it is not necessary to find exact solutions of the intermediate ground-state problems. In fact, even if the Ansatz does not fully converge to the instantaneous variational ground-state at some intermediate step, this approximate solution will still serve as a valid warm start for the optimization problem in Eq. (12) at the next annealing step. Previous numerical errors can be compensated when re-optimizing the Ansatz, a feature that does not extend to real dynamics on a quantum annealer or real-time TDVP simulations. Guided by this observation, we recognize a trade-off in the choice of N_A and N_U , as the total computational cost scales linearly with their product. Rather than fully optimizing the variational parameters at each annealing step, it is often more efficient to increase the number of annealing steps N_A while performing only a few parameter updates N_U per step. In practice, even a single update per step ($N_U = 1$) can be sufficient to reach the final target ground state. To ensure that any residual excited-state components are suppressed at $j = N_A$, the variational parameters can be fine-tuned with a few additional optimization epochs as in Eq. (29).

Inclusion of a catalyst The classical simulation of approximate adiabatic dynamics enables full flexibility in choosing Hamiltonian terms. We test the introduction of a catalyst H_C , which is an auxiliary Hamiltonian term vanishing at both $s = 0$ and $s = 1$:

$$H(s) = s H_T + A(1 - s) H_0 + B s(1 - s) H_C, \quad (30)$$

where H_0 is given by Eq. (9), and $A > 0$ and B are two tunable real hyper-parameters. By reshaping the instantaneous energy landscape, catalysts can convert a first-order phase transition encountered in real-time dynamics – with an exponentially small spectral gap – into a higher-order transition with a polynomially closing gap [58, 59]. Similarly, it may help reduce diabatic errors during variational optimization in a VQA setting. In this work, we select

$$H_C = \sum_{j=1}^N \hat{\sigma}_j^y \quad (31)$$

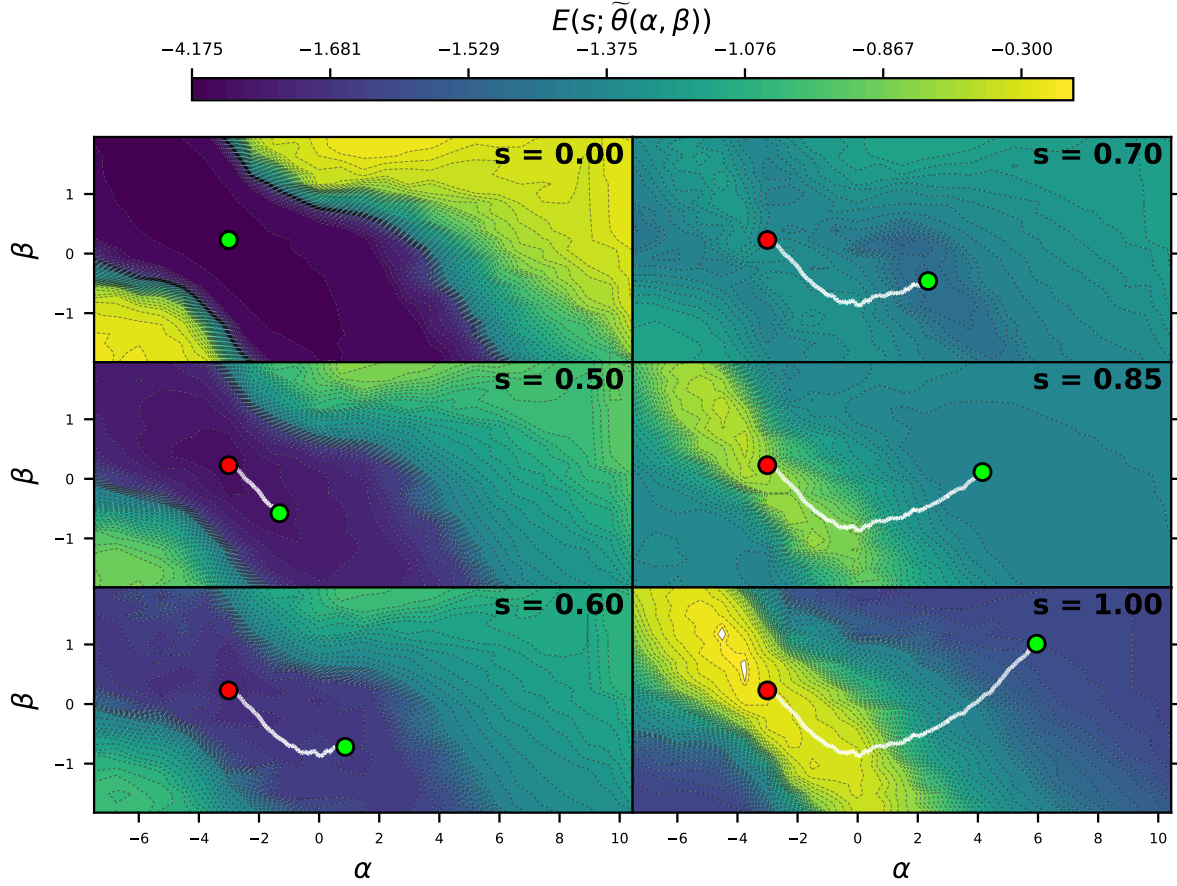


Figure 1: Visualization of the NQA trajectory for a 4-sites Ising model with all-to-all random interaction. Each subplot shows the parameters trajectory and the evolution of the energy landscape in Eq. (8) at different points in the annealing schedule. Here, the white line is the projection of the trajectory of the parameters on a plane obtained by a dimensionality reduction to the first two principal components (α, β) obtained by applying PCA to the parameter trajectory. Details are reported in appendix... Parameter trajectory with red and green dots representing the starting and current parameters, the background is the instantaneous energy landscape with the corresponding reconstructed parameters (how do you reconstruct it?)

as a catalyst, which is a non-stoquastic term that cannot be implemented on current quantum annealers [3].

8 Numerical results

In this Section, we benchmark NQA on hard classical and quantum ground states, demonstrating its effectiveness for long-range models with frustration and NP-complete real-world problems. All the data and code are available at [?].

8.1 Sherrington–Kirkpatrick model

We test our methods on the Sherrington–Kirkpatrick (SK) model, a standard benchmark for combinatorial optimization algorithms. It is defined by N spin-1/2 degrees of freedom with all-to-all interactions whose coupling constants are sampled independently from a normal distribution. The Hamiltonian of the model reads

$$H = \sum_{i < j} J_{ij} S_i S_j, \text{ with } J_{ij} \sim \mathcal{N}(0, 1/N). \quad (32)$$

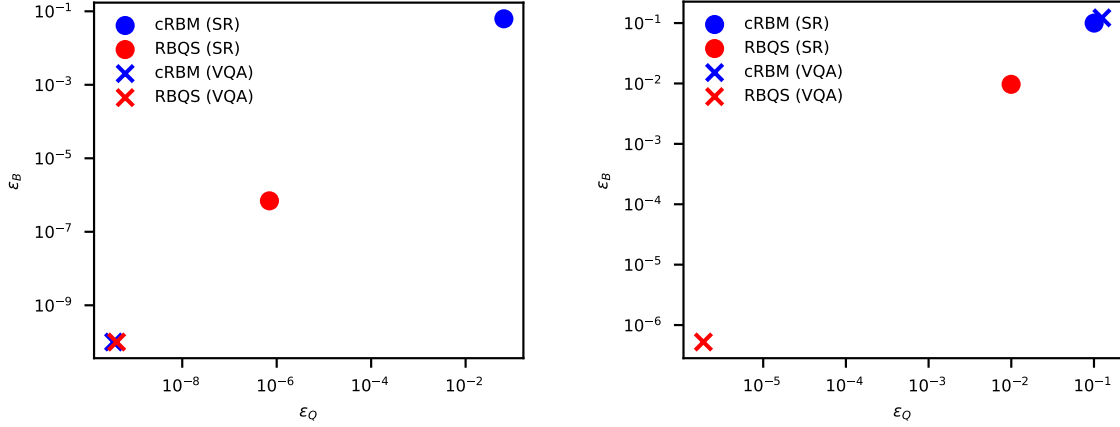


Figure 2: Comparison of the typical relative errors on the realizations of the SK model with $N = 100$ (left) and $N = 200$ (right). Each point corresponds to a different Ansatz and optimization protocol combinations.

We prepared 10 realizations of the SK model with $N = 100$ spins and 10 realizations with $N = 200$ to benchmark our set-up. Reference ground-state energies for these realizations were computed via 1000 runs of simulated classical annealing using OpenJij [60]. The spin-glass server [61], a resource commonly used to obtain exact ground states from comparatively small-sized instances, only managed to solve 8 out of the 10 instances with $N = 100$ spins, reaching the same ground state energy as the simulated annealing. These instances represent NP-hard problems where traditional solvers already struggle for $N \approx 100$, and $N = 200$ pushes into a regime requiring more scalable methods. Many combinatorial optimization problems that are omnipresent in science and industry can be mapped to similar spin models (e.g.: traveling sales man, graph coloring, JSSP). Moreover, realizations of the SK model lack geometrical structures that can be leveraged to implement efficient inductive biases in a NQS, as in the case of CNNs and convolutional transformers for translationally invariant models. For each run, we evaluate two relative energy error metrics to quantify solution quality: the final quantum state error $\epsilon_Q \equiv (\langle H \rangle - E_0)/|E_0|$, and the energy error of the best classical state encountered during the simulation $\epsilon_B \equiv (E_B - E_0)/|E_0|$. For a classical spin glass, $\epsilon_B = 0$ is a clear success criterion (ground state found as a configuration), even if the final wave function has not fully collapsed onto that configuration.

We compare the performance of real-valued cRBM and RBQS on the benchmark problems, when optimized via SR and NQA. This approach is justified by the fact that the QA Hamiltonian in Eq. (10) is stoquastic for a standard driver and target Hamiltonian in Eq. (9) and Eq. (11), respectively, and the ground state can consequently be described by a real-valued wave function. Sampling is performed with block Gibbs updates or single spin-flip Metropolis-Hasting chains that are kept persistent across annealing steps. To ensure a fair comparison we conduct a short hyperparameter optimization with Optuna [62], allocating the same number of trials and the same wall-clock budget per trial to each (algorithm, Ansatz) pair. Here, we report the best-found settings, additional details are reported in App. ???. To aggregate results over multiple instances, we report typical (geometric mean) errors

$$[x]_{\text{typ}} \equiv \exp(\langle \log x \rangle), \quad (33)$$

$\epsilon_Q = [\epsilon_Q]_{\text{typ}}$ and $\epsilon_B = [\epsilon_B]_{\text{typ}}$, using the exponential of the average log-error (with a 10^{-10} floor for numerical stability). Figures 3, shows the histograms for the residual energies. We find that our optimization algorithm always outperforms SR regardless of the variational Ansatz used. While the instances with $N = 100$ spins are all solved by both Ansätze when optimized via NQA, only the combination of RBQS with NQA manages to solve instances with $N = 200$ spins, solving half of them.

Performance on the $N = 200$ instances can be improved with the inclusion of a catalyst, and complex-valued parameters to describe the non-stoquastic instantaneous ground state. Figures 4,

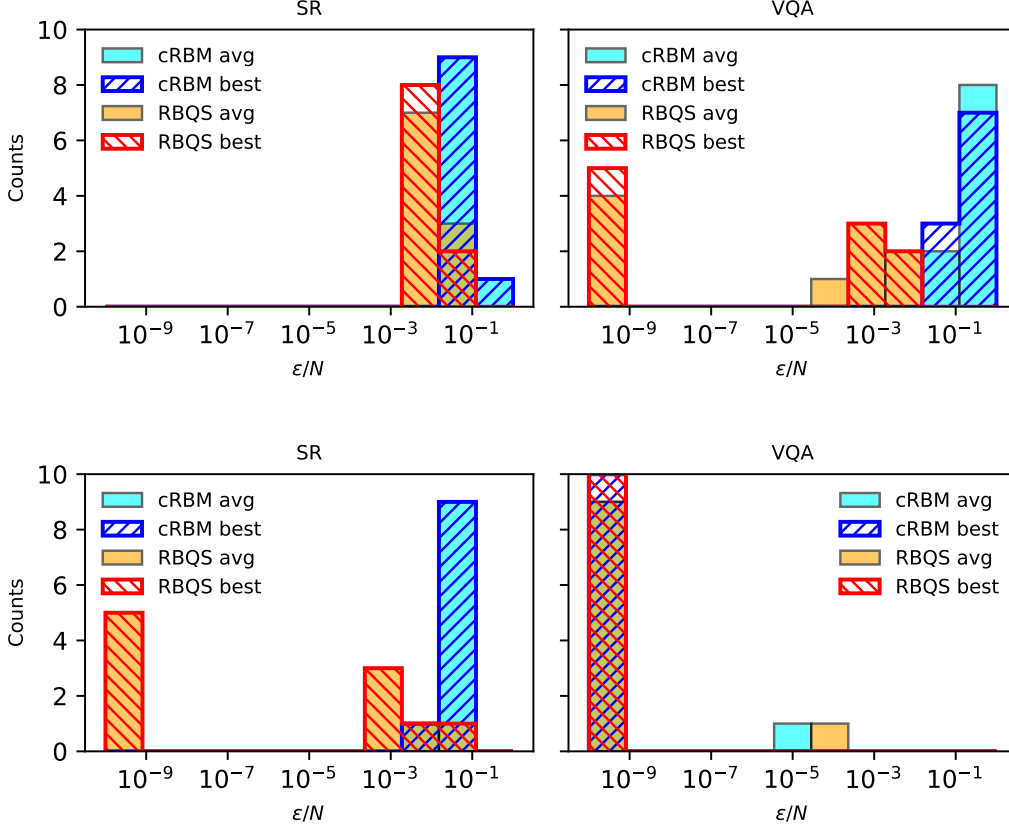


Figure 3: Residual energies histogram for each combination of variational Ansatz and optimization strategy for the 10 realization of the SK model with $N = 100$ (top panels) and $N = 200$ (bottom panels) spins. Left panel shows the comparison of the final energies of real parameterized cRBM and RBQS Ansätze when optimized via standard SR. Right panel showcases the same comparison when the Ansätze are optimized via NQA. Solid bars represent the average energy of the final quantum state, and dashed bars represent the lowest classical energy encountered during the simulation. Leftmost bars corresponds to instances solved exactly. Can we move this to the appendix?

shows the histograms for the residual energies. Finally, including a second hidden layer, and optimizing the DBQS via VQA yields the best performing algorithm, as shown in Figure 5.

8.2 Transverse field Sherrington–Kirkpatrick model

To benchmark our set-up on quantum spin glasses, we consider the Sherrington-Kirkpatrick model in a transverse field, i. e.

$$H = \sum_{i < j} J_{ij} S_i^z S_j^z + g \sum_i S_i^x, \text{ with } J_{ij} \sim \mathcal{N}(0, 1/N). \quad (34)$$

Specifically, we consider the same realizations of the SK model used in the previous section. This will tell us how well our variational scheme is able to represent the intermediate stages of the annealing. Here we have no external reference to compare with, therefore we fall back to internal consistency checks to assert convergence to the ground state. In the presence of transverse field in the classical SK model, some degeneracies between nearly equivalent spin glass states are lifted due to the quantum fluctuations. So in the limit of weak transverse field, quantum tunneling between classically degenerate configurations results in an effective renormalization of couplings that favors spin-glass order. Along with that a simple variational calculation tells us that the ground state energy of the quantum model has to be lower than its classical counterpart.

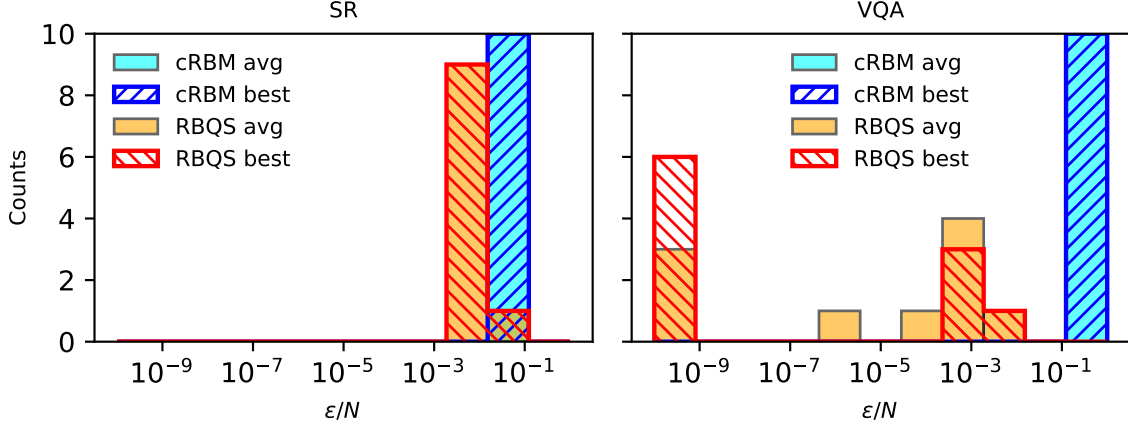


Figure 4: Residual energies histogram for each combination of variational Ansatz and optimization strategy for the 10 realization of the SK model with $N = 200$ spins. Left panel shows the comparison of the final energies of complex parameterized cRBM and RBQS Ansätze when optimized via standard SR. Right panel showcases the same comparison when the Ansätze are optimized via VQA, with the inclusion of a catalyst. In both panels, the solid bars represent the average energy of the final quantum state, and the dashed bars represent the lowest classical energy encountered during the simulation. Leftmost bars corresponds to instances solved exactly.

In our paper we use the same 10 instances of 100 sites for the purpose of classical and quantum SK model benchmarking. We choose a small transverse field (0.1 in our case) to make sure we are deep in the spin glass phase. Due to introduction of quantum correlations it is usually harder for the RBQS ansatz to capture the ground state of the QSK model compared to the classical SK model. So we usually need larger network and more number of annealing steps for convergence.

We use the Optuna for hyperparameter optimization of the Quantum SK model as well. In all our optimization we fix a few hyperparameters through our educated guess (from internal checks and by Optuna optimization of all the hyperparameters, fixing some hyperparameter makes the convergence to the desired set of hyperparameters much more time efficient) and optimize the important hyperparameters which tend to affect the model most. In this process of optimizing the hyperparameters we also capture some of the low lying states of the transverse field SK model. From the figure of fluctuations corresponding to the target energy we can see the low lying states are actually eigenstates of the transverse field SK model.

Convergence with respect to the number of variational parameters. Energy variance expected to be vanishing and energy variance extrapolation.

8.3 JSSP

To benchmark our set-up on large scale spin glasses we turn to the Ising formulation of the Job Shop Scheduling Problem. The resulting Ising spin glass has zero ground-state energy by construction, where the energy represents the number of violated constraints in the schedule, therefore providing a benchmark to see how far the algorithm can be pushed. We found exact solutions for 9 instances of JSSP with number of spins 88, 104, 164, 220, 295, 305, 345, 372, 725. [A figure showcasing a decoded schedule here would be nice.](#)

9 Discussion

- Some form of universal approximation theorem would give the algorithm a better theoretical motivation. Note that for the case of classical (or stoquastic) hamiltonian that can be described by real amplitudes we have the universal approx theorem for free thanks to the mapping to boltzmann machines.

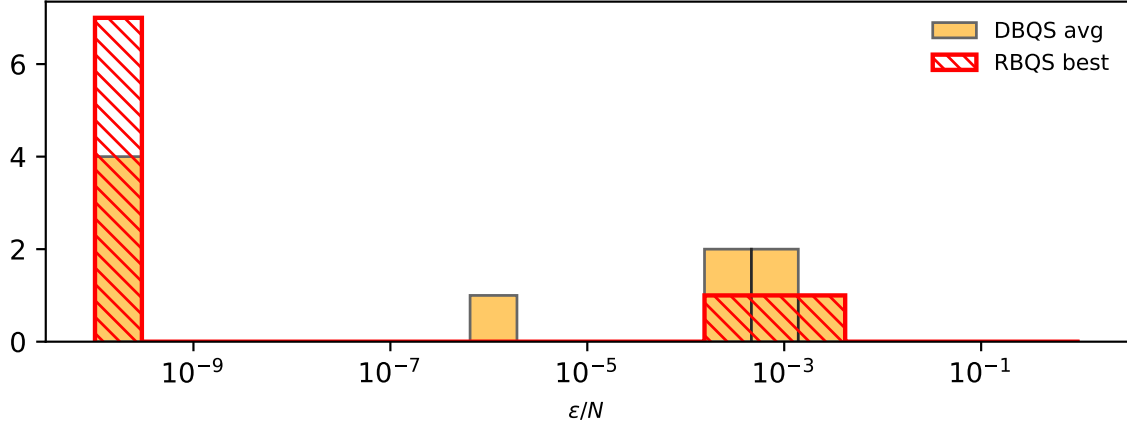


Figure 5: Residual energies histogram for DBQS Ansatzes optimized via NQA including a catalyst for the 10 realization of the SK model with $N = 200$ spins.

- It is possible to make the RBQS and DBQS convolutional just by replacing the matrix products with convolutions with small kernels, as already done by the ML community.

10 Future works

- Try a parallel tempering version of our algorithm: exchange configurations among Markov chains at different values of the coupling to avoid mode collapse more effectively?
- Going beyond Quadratic hamiltonians? In theory the algorithm should work for any binary optimization problem.
- Solving ground-states of Quantum Many Body problems (the ones we did here are all classical target Hamiltonian)
- Can Boltzmann Quantum States be extended to study fermionic ground-states by simply including a Slater determinant? (This logically goes after true quantum gs problems)
- real time dynamics with history states, see section about Feynman-Kitaev Hamiltonian in for a quick overview <https://journals.aps.org/prresearch/pdf/10.1103/PhysRevResearch.4.043161>
- Use and test transformers for both NQA and SR

A Algorithmic details on NQS optimization for ground-state search

A.1 Natural gradients and advanced optimizers

In Section 2, we introduced the real and imaginary-time TDVP and defined the Fisher information matrix. A noteworthy observation is that imaginary-time TDVP is formally equivalent to the optimization of the cost function in Eq. (8) via natural gradient descent, a seminal algorithm known as Stochastic Reconfiguration (SR) [16], recently generalized by min-SR [2]. This equivalence follows from the definition of natural gradient, obtained by applying the inverse Fisher information matrix to the standard gradient:

$$\mathbf{g}_\theta = S^{-1}(\theta) \nabla_\theta. \quad (35)$$

Indeed, a first-order numerical integration of the imaginary-time TDVP equations leads to an iterative parameter update

$$\theta(\tau + \delta\tau) = \theta(\tau) - \delta\tau S^{-1}(\theta(\tau)) F(\theta(\tau)), \quad (36)$$

which corresponds to a Natural Gradient Descent step with step size $\delta\tau$ since

$$\mathbf{g}_\theta E(\boldsymbol{\theta}) = S^{-1}(\boldsymbol{\theta}) \nabla_\theta E(\boldsymbol{\theta}) = S^{-1}(\boldsymbol{\theta}) F(\boldsymbol{\theta}). \quad (37)$$

As briefly mentioned in Section 2, natural gradients can be combined with advanced stochastic optimizers developed for deep learning. While this strategy has not been explored in the NQS literature to our knowledge, it has been studied in the broader machine learning community. For example, works such as [63, 64] precondition the stochastic gradient with an (approximate) inverse Fisher information matrix and apply momentum before updating parameters. Ref. [65] further proposes a unified framework for accelerated natural gradient descent.

In practice, an optimizer maintains an auxiliary state α , and given the (natural) gradient, returns both the parameter update and the new auxiliary state. As stated in Section 7, in this work, we combine SR with momentum, where the auxiliary state is a velocity vector storing an exponentially decaying average of past (natural) gradients.

Algorithm 1 presents a high-level overview of NQS optimization for ground-state search, combining natural gradients with an advanced optimizer.

Algorithm 1 NQS optimization for ground-state search

Require: Number of training epochs N_ϵ ; Optimizer $\text{opt}(\mathbf{g}, \alpha)$; Variational Ansatz $\psi(\mathbf{x}; \boldsymbol{\theta})$.

Ensure: Optimized parameters $\boldsymbol{\theta}$

- 1: Initialize parameters $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$
 - 2: Initialize auxiliary state of the optimizer $\alpha \leftarrow \alpha_0$
 - 3: **for** $n = 0$ **to** $N_\epsilon - 1$ **do**
 - 4: Sample a set of configurations $\{\mathbf{x}\}$ from $p(\mathbf{x}) \propto |\psi(\mathbf{x}; \boldsymbol{\theta})|^2$
 - 5: Compute the (natural) gradient $\mathbf{g} \leftarrow \mathbf{g}_\theta E(\boldsymbol{\theta})$
 - 6: Compute the updates for the parameters and the new optimizer state $(\delta\boldsymbol{\theta}, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$
 - 7: Update the variational parameters $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \delta\boldsymbol{\theta}$
 - 8: **end for**
 - 9: **return** $\boldsymbol{\theta}$
-

A.2 Variational Quantum Annealing pseudo-code

As explained in Section 4, an alternative approach to ground-state optimization with NQS relies on interpolating from an *easy* to a *hard* regime by tuning a parameter in the Hamiltonian. The resulting high-level implementation of VQA for optimizing an NQS architecture is illustrated in Algorithm 2. We note that, in Ref. [4], the parametric Hamiltonian is defined slightly differently compared to Eq. (10), precisely as $H(s) = H_T + (1-s)H_0$, with $H_0 = -B^x \sum_i \sigma_i^x$ and $B^x \gg 1$. Although Ref. [4] uses Adam and simple gradients to train RNN wavefunctions, it is possible to extend this approach to other NQS architectures and to natural gradients.

Algorithm 2 Variational Quantum Annealing for ground-state search

Require: Number of warm-up steps N_W ; Number of annealing steps N_A ; Number of updates per annealing step N_U ; Optimizer $\text{opt}(\mathbf{g}, \alpha)$ (Adam and simple gradients in [4]); Variational Ansatz $\psi(\mathbf{x}; \boldsymbol{\theta})$ (RNN in [4]).

Ensure: Optimized parameters $\boldsymbol{\theta}$

```
1: Initialize parameters  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$ 
2: Initialize auxiliary state of the optimizer  $\alpha \leftarrow \alpha_0$ 
3:  $H(s) \leftarrow H_0$ 
4: for  $n = 0$  to  $N_W - 1$  do
5:   Sample a set of configurations  $\{\mathbf{x}\}$  from  $p(\mathbf{x}) \propto |\psi(\mathbf{x}; \boldsymbol{\theta})|^2$ 
6:   Compute (natural) gradients of the variational energy  $\mathbf{g} \leftarrow \nabla_{\boldsymbol{\theta}} E(s; \boldsymbol{\theta})$ 
7:   Compute the updates for the parameters and the new optimizer state  $(\delta\boldsymbol{\theta}, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$ 
8:   Update the variational parameters  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \delta\boldsymbol{\theta}$ 
9: end for
10: for  $j = 1$  to  $N_A$  do
11:    $s \leftarrow j/N_A$ 
12:    $H(s) \leftarrow sH_T + (1-s)H_0$ 
13:   for  $n = 0$  to  $N_U - 1$  do
14:     Sample a set of configurations  $\{\mathbf{x}\}$  from  $p(\mathbf{x}) \propto |\psi(\mathbf{x}; \boldsymbol{\theta})|^2$ 
15:     Compute (natural) gradients of the variational energy  $\mathbf{g} \leftarrow \nabla_{\boldsymbol{\theta}} E(s; \boldsymbol{\theta})$ 
16:     Compute the updates for the parameters and the new optimizer state  $(\delta\boldsymbol{\theta}, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$ 
17:     Update the variational parameters  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \delta\boldsymbol{\theta}$ 
18:   end for
19: end for
20: return  $\boldsymbol{\theta}$ 
```

B Standard analytical results on Boltzmann Machines

In Section 5, we introduced the RBM as the simplest efficiently trainable version of generic BMs. Starting from Eq. (15), the joint probability function over visible and hidden variables reads

$$p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta}) = Z^{-1} \exp\{\mathbf{v}^T W \mathbf{h} + \mathbf{b}^T \mathbf{v} + \mathbf{c}^T \mathbf{h}\}, \quad Z = \sum_{\mathbf{v}', \mathbf{h}'} \exp\{-E(\mathbf{v}', \mathbf{h}'; \boldsymbol{\theta})\}. \quad (38)$$

This partition function Z has been proven to be intractable [66], implying that the explicit computation of a normalized marginal probability distribution for the visible units is also intractable. Nonetheless, it is possible to exactly trace out the hidden variables to obtain the unnormalized marginal distribution

$$p_{\text{RBM}}(\mathbf{v}; \boldsymbol{\theta}) = \sum_{\mathbf{h}} p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta}) = Z^{-1} \exp\{\mathbf{b}^T \mathbf{v}\} \prod_{j=0}^{N_h} \cosh\left\{c_j + \sum_{i=0}^{N_v} v_i W_{ij}\right\}. \quad (39)$$

The inability to compute the exact partition function does not constitute a serious limitation, as multiple methods have been devised to approximate it [46], and its explicit computation is often unnecessary in practice: Monte Carlo methods circumvent the problem of computing the partition function by computing unweighted averages over the samples drawn from the model.

The key property of RBMs that enables efficient sampling and training stems from the bipartite structure of their interaction graph: conditioned on the hidden units, the visible units are mutually independent, and vice versa. This allows us to explicitly compute the normalized conditional probabilities

$$\begin{aligned} p(\mathbf{v}|\mathbf{h}; \boldsymbol{\theta}) &= \frac{p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})}{p(\mathbf{h}; \boldsymbol{\theta})} = \frac{1}{Z} \frac{\exp\{\mathbf{c}^T \mathbf{h}\}}{p(\mathbf{h}; \boldsymbol{\theta})} \exp\{\mathbf{b}^T \mathbf{v} + \mathbf{v}^T W \mathbf{h}\} = \frac{1}{Z'} \exp\{\mathbf{b}^T \mathbf{v} + \mathbf{v}^T W \mathbf{h}\} \\ &= \frac{1}{Z'} \prod_{i=1}^{N_v} \exp\left\{\left(b_i + \sum_{j=1}^{N_h} W_{ij} h_j\right) v_i\right\} \equiv \frac{1}{Z'} \prod_{i=1}^{N_v} \tilde{p}(v_i|\mathbf{h}; \boldsymbol{\theta}). \end{aligned} \quad (40)$$

Here, the hidden variables are treated as constants, as we are conditioning on their values. Hence, all terms not explicitly depending on visible units are included in a new constant Z' . The conditional probability distribution is factorized over the visible units, and the single terms can be easily normalized since computing the partition function in this case involves a sum over two configurations only. The normalized conditional probabilities read:

$$\begin{aligned}
p(v_i|\mathbf{h};\boldsymbol{\theta}) &= \frac{\tilde{p}(v_i|\mathbf{h};\boldsymbol{\theta})}{\tilde{p}(v_i=+1|\mathbf{h};\boldsymbol{\theta}) + \tilde{p}(v_i=-1|\mathbf{h};\boldsymbol{\theta})} \\
&= \frac{\exp\left\{\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)v_i\right\}}{\exp\left\{\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right\} + \exp\left\{-\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right\}} \\
&= \exp\left\{\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)(v_i - 1)\right\} \sigma\left[2\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right],
\end{aligned} \tag{41}$$

where $\sigma(x)$ denotes the logistic sigmoid function. The normalized probability for the i -th visible unit to have value $+1$, conditioned on all hidden variables, simplifies to

$$p(v_i=+1|\mathbf{h};\boldsymbol{\theta}) = \sigma\left[2\left(b_i + \sum_{j=1}^{N_h} W_{ij}h_j\right)\right], \tag{42}$$

and a similar expression holds for the value -1 . Since the difference between visible and hidden units is only formal, the conditional probability for the hidden units given the visible ones can be computed in the same way:

$$p(h_j=+1|\mathbf{v};\boldsymbol{\theta}) = \sigma\left[2\left(c_j + \sum_{i=1}^{N_v} W_{ij}v_i\right)\right]. \tag{43}$$

The mutual independence of visible units conditioned on hidden units, and vice versa, enables efficient block Gibbs sampling of the joint probability function $p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})$. By starting from a random configuration, a Markov chain can be constructed with the update rule that, at each iteration, all visible units are simultaneously sampled conditioned on the hidden units, and then all hidden units are sampled conditioned on the visible units. Moreover, these samples can be drawn via exact sampling of the conditional probability in Eqs. (42) and (43). Such an alternating conditional sampling efficiently generates samples from the joint probability distribution $p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})$ [67, 46], which are necessary both during training and inference.

During training, the RBM parameters $\boldsymbol{\theta}$ are typically optimized by minimizing the negative log-likelihood using variants of gradient descent. The gradient of such objective function can be expressed in terms of expectation values and two-point correlations between hidden and visible units [68], which are efficiently estimated via Gibbs sampling. At the inference stage, the trained model would be used to compute the expectation values of functions of the visible units $f(\mathbf{v})$ over the marginal probability distribution $p_{\text{RBM}}(\mathbf{v}; \boldsymbol{\theta})$ in Eq. (39), which cannot be explicitly normalized. This expectation value, denoted as $\langle \dots \rangle_{\text{RBM}}$, is equal to the expectation value of the same function over the joint probability $p(\mathbf{v}, \mathbf{h}; \boldsymbol{\theta})$:

$$\langle f(\mathbf{v}) \rangle_{\text{RBM}} = \frac{\sum_{\mathbf{v}} p(\mathbf{v}) f(\mathbf{v})}{\sum_{\mathbf{v}} p(\mathbf{v})} = \frac{\sum_{\mathbf{v}, \mathbf{h}} p(\mathbf{v}, \mathbf{h}) f(\mathbf{v})}{\sum_{\mathbf{v}, \mathbf{h}} p(\mathbf{v}, \mathbf{h})} = \langle f(\mathbf{v}) \rangle_{p(\mathbf{v}, \mathbf{h})}. \tag{44}$$

This allows us to efficiently estimate it by performing block Gibbs sampling and using the normalized conditional probability functions in Eqs. (42) and (43).

The more general case of DBMs in Eq. (16) is similarly treated. The joint probability function is given by

$$p(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{Z} \exp\left\{\sum_{l=0}^{N_L-1} \mathbf{x}^{(l)T} W^{(l)} \mathbf{x}^{(l+1)} + \sum_{l=0}^{N_L} \mathbf{b}^{(l)T} \mathbf{x}^{(l)}\right\}, \quad Z = \sum_{\mathbf{x}'} \exp\{-E(\mathbf{x}'; \boldsymbol{\theta})\}. \tag{45}$$

The computation of the marginal probability function for the visible units, as done for the RBM in Eq. (39), is now intractable even in its unnormalized version. Nonetheless, similarly to the case of

RBM, the variables can be rearranged into a bipartite graph, with odd layers on one side and even layers on the other. This immediately implies that when we condition on the units in the even layers, the variables in the odd layers become independent, and vice versa. More precisely, it is enough to condition on the adjacent layers, and the explicit computation of the normalized conditional probability leads to

$$p(x_i^{(l)} = +1 | \mathbf{x}^{(l-1)}, \mathbf{x}^{(l+1)}; \boldsymbol{\theta}) = \sigma \left[2 \left(\sum_j W_{ji}^{(l-1)} x_j^{(l-1)} + \sum_j W_{ij}^{(l)} x_j^{(l+1)} + b_i^{(l)} \right) \right], \quad (46)$$

with slightly different expressions for the first and last layers. Analogously to the RBM case, this conditional independence enables efficient block Gibbs sampling by updating all units in only two iterations. Firstly, one can update all the units in the even layers (including the $l = 0$ visible layer) as a block, conditioned on the units in the odd layers, and, secondly, all the hidden units in the odd layers conditioned on the units in the even ones. The samples can be drawn via exact sampling of the conditional probability in Eq. (46).

C Previous quantum extensions of Boltzmann Machines

Several extensions of BMs and specifically RBMs to quantum spin systems have been previously proposed. For quantum systems, the fundamental object is not the probability function, but rather the wavefunction, which is a complex-valued function representing the probability amplitudes.

C.1 pmRBM and cRBM

A conceptually straightforward extension proposed in Ref. [69] and known as pmRBM uses two independent RBMs to model the phase and the modulus of the wavefunction. The wavefunction is expressed in the spin eigenbasis, which is commonly associated with the eigenstates of the Pauli-Z operators:

$$\psi(\mathbf{x}; \boldsymbol{\theta}) = \sqrt{p_{\text{RBM}}^m(\mathbf{x}; \boldsymbol{\theta}_m)} \exp\{i p_{\text{RBM}}^p(\mathbf{x}; \boldsymbol{\theta}_p)\}, \quad (47)$$

where $p_{\text{RBM}}(\mathbf{x}; \boldsymbol{\theta})$ denotes the RBM unnormalized marginal probability distribution (see Appendix B). Here, $\boldsymbol{\theta} = (\boldsymbol{\theta}_m, \boldsymbol{\theta}_p)$ are real parameters, and $\mathbf{x}$ is a collective label for a spin eigenstate in the Pauli-Z eigenbasis. Current literature suggests that the artificial separation of modulus and phase information is suboptimal, making the learning problem more challenging, regardless of the specific ANN model used. Almost all recent NQS applications [13] utilize ~~an~~ ANNs parameterized by complex variational parameters to output a complex number representing the wavefunction amplitude. This approach was proposed in the context of RBMs in Ref. [1], which is based on the idea of promoting the probability distribution defined by an RBM to quantum amplitudes by introducing complex parameters. The resulting Ansatz, commonly referred to as cRBM, reads

$$\psi(\mathbf{x}; \boldsymbol{\theta}) = \exp \left\{ \sum_{i=0}^{N_v} b_i x_i \right\} \prod_{j=0}^{N_h} \cosh \left\{ c_j + \sum_{i=0}^{N_v} x_i W_{ij} \right\}, \quad (48)$$

which is formally equivalent to the unnormalized marginal probability distribution of an RBM (see Appendix B), with the only difference that $\boldsymbol{\theta} = (W, \mathbf{b}, \mathbf{c})$ are now complex parameters.

In combination with the SR algorithm, this Ansatz has been successfully used to find a variety of quantum ground-states with remarkable precision [1, 70, 71, 72] and in the context of NQA in Ref. [43]. However, the possibility of efficient block Gibbs sampling is lost in the cRBM, so one must utilize Metropolis-Hastings sampling with local update rules, which has significantly longer autocorrelation times, making it slower and more computationally expensive.

C.2 DBM wavefunctions

A possible quantum extension of DBMs based on the same principle of the cRBM was introduced in Ref. [8] and reads

$$\psi(\mathbf{x}; \boldsymbol{\theta}) = \sum_{\mathbf{x}^{(l)}, l \geq 0} \exp \left\{ \sum_{l=0}^{N_L-1} \mathbf{x}^{(l)T} W^{(l)} \mathbf{x}^{(l+1)} + \sum_{l=0}^{N_L} \mathbf{b}^{(l)T} \mathbf{x}^{(l)} \right\}. \quad (49)$$

The original reference also provides an interesting constructive approach that does not require a variational optimization of the DBM parameters. However, due to the intractability of the summation in Eq. (49) and the restrictive conditions under which efficient block Gibbs sampling of the model is possible, this variational Ansatz has only been applied to small-scale benchmarks.

C.3 Quantum Boltzmann Machines

Another possible extension, dubbed Quantum Boltzmann Machine (QBM), was proposed in Ref. [68]. The units of the BM are promoted to quantum spins, and the state of the system is described as the exponential of a transverse-field Ising Hamiltonian

$$H_{\text{QBM}} = - \sum_a \Gamma_a \sigma_a^x - \sum_a b_a \sigma_a^z - \sum_{ab} w_{ab} \sigma_a^z \sigma_b^z \quad (50)$$

yielding the density matrix operator

$$\rho = \frac{1}{Z} \exp\{-H_{\text{QBM}}\}. \quad (51)$$

The presence of non-commuting terms poses significant challenges in training the model, as detailed in the original reference [68], which applies this framework to address classical machine learning tasks. Later, Ref. [73] used a simplified version of QBMs without hidden units to learn the ground-states of quantum systems through a hybrid quantum-classical method inspired by the variational quantum eigensolver.

D Analytical results for Boltzmann Quantum States

In this Section, we report additional results on BQS and their relationship to the cRBM introduced in [8] and described in Appendix C. We adopt the same notation as in Section 6.

D.1 Ground-state structure of the extended Hilbert space and expectation values

The BQS formalism describes both visible and hidden spins as quantum spins in an extended Hilbert space $\mathcal{H}_{vh} = \mathcal{H}_v \otimes \mathcal{H}_h$. Although the two Hamiltonians H_v and $H_{vh} = H_v \otimes \mathbb{I}_h$ share the same spectrum, the ground-state of the extended system can have a non-trivial entanglement structure. Specifically, if H_v has a degenerate ground-state, then there exist ground-states of H_{vh} that are entangled across the visible/hidden partition, rather than simple tensor products of two pure states.

Let H_v be a Hamiltonian with a d -fold degenerate ground-state, let $\{|\Psi_v^a\rangle\}_{a=1}^d$ be a set of orthonormal ground states, and let $\{|\Phi_v^b\rangle\}_{b=1}^{\dim \mathcal{H}_v - d}$ be a set of orthonormal higher energy eigenstates so that $\{|\Psi_v^a\rangle, |\Phi_v^b\rangle\}$ forms a complete orthonormal basis of $\mathcal{H}_v$. Then, any state $|\psi_{vh}\rangle$ of the extended system can be expanded as

$$|\psi_{vh}(G, E)\rangle = \sum_{a, \mathbf{h}} G_{a\mathbf{h}} |\Psi_v^a\rangle \otimes |\mathbf{h}\rangle + \sum_{b, \mathbf{h}} E_{b\mathbf{h}} |\Phi_v^b\rangle \otimes |\mathbf{h}\rangle, \quad (52)$$

and its energy expectation value is

$$\frac{\langle \psi_{vh}(G, E) | H_{vh} | \psi_{vh}(G, E) \rangle}{\langle \psi_{vh}(G, E) | \psi_{vh}(G, E) \rangle} = \frac{\sum_{a, \mathbf{h}} |G_{a\mathbf{h}}|^2 \langle \Psi_v^a | H_v | \Psi_v^a \rangle + \sum_{b, \mathbf{h}} |E_{b\mathbf{h}}|^2 \langle \Phi_v^b | H_v | \Phi_v^b \rangle}{\sum_{a, \mathbf{h}} |G_{a\mathbf{h}}|^2 + \sum_{b, \mathbf{h}} |E_{b\mathbf{h}}|^2}. \quad (53)$$

Hence, by the variational principle, the ground states of H_{vh} take the form

$$|\Psi_{vh}(G)\rangle = \sum_{a, \mathbf{h}} G_{a\mathbf{h}} |\Psi_v^a\rangle \otimes |\mathbf{h}\rangle, \quad (54)$$

and variational energy minimization in the extended Hilbert space can generally yield entangled states that does not decompose as the tensor product of a state of the visible spins and a state of the hidden ones, regardless of the variational Ansatz used. Consequently, it is not generally possible to trace out

the hidden degrees of freedom to obtain a pure state for the visible spins. While it is theoretically possible to obtain a reduced density matrix for the visible spins to describe the mixed state in which they are found

$$\rho_v = \sum_{\mathbf{v}, \mathbf{v}', \mathbf{h}} \psi(\mathbf{v}, \mathbf{h}) \overline{\psi(\mathbf{v}', \mathbf{h})} |\mathbf{v}'\rangle \langle \mathbf{v}|, \quad (55)$$

the summation over $\mathbf{h}$ required to compute the matrix elements is tractable in the case of RBQS Ansätze, and an analytical expression can be obtained for the reduced density operator

$$\rho_v^{\text{RBQS}} = \sum_{\mathbf{v}, \mathbf{v}'} \exp \left\{ \left[\sum_i b_i v_i - \bar{b}_i v_i \right] \right\} \prod_j \cosh \left\{ \sum_i (W_{ij} v_i - \bar{W}_{ij} v'_i) + c_j - \bar{c}_j \right\} |\mathbf{v}'\rangle \langle \mathbf{v}|. \quad (56)$$

Similarly to the classical case, the summation is untractable for general BQS Ansätze, and it won't be possible to obtain a closed formula for the density operator.

It is however possible to obtain the ground-state expectation value of any operator O_v acting solely on visible spins from the ground state of the extended system by computing the expectation value of the operator $O_v \otimes \mathbb{I}_h$. Expectation values on a general ground state of the extended system are more conveniently computed using the Schmidt decomposition [?]: given any state of the form of Eq. (54), let $(U, \tilde{G}, V)$ be the singular value decomposition of G , $G = U \tilde{G} V^\dagger$, then we can write $|\Psi\rangle_{vh}$ as

$$|\Psi_{vh}(G)\rangle = \sum_n \tilde{G}_n |\tilde{\Psi}_v^n\rangle \otimes |\tilde{\mathbf{h}}^n\rangle, \quad (57)$$

where $|\tilde{\Psi}_v^n\rangle = U |\psi_v^a\rangle$, $|\tilde{\mathbf{h}}^n\rangle = V^\dagger |\mathbf{h}\rangle$, and $\tilde{G}_n = \tilde{G}_{nn}$. Using such expansion immediately expose that the operator $O_v \otimes \mathbb{I}_h$ on a general ground state of the extended system is

$$\frac{\langle \Psi_{vh}(G) | O_v \otimes \mathbb{I}_h | \Psi_{vh}(G) \rangle}{\langle \Psi_{vh}(G) | \Psi_{vh}(G) \rangle} = \frac{\sum_{n,m} \tilde{G}_n G_m \langle \tilde{\Psi}_v^n | O_v | \tilde{\Psi}_v^m \rangle \langle \tilde{\mathbf{h}}^n | \tilde{\mathbf{h}}^m \rangle}{\sum_{a,\mathbf{h}} |G_n|^2} = \sum_n p_n \langle \tilde{\Psi}_v^n | O_v | \tilde{\Psi}_v^n \rangle, \quad (58)$$

where $p_n = |G_n|^2 / (\sum_m |G_m|^2)$. Unsurprisingly, this is the same result that would be found by first tracing out the hidden units to obtain a reduced density matrix for the visible units ρ_v , and then computing $\langle O_v \rangle = \text{tr}\{\rho_v O_v\}$.

In practical application, after a variational optimization, a DBQS converges to a close approximation of some ground state of the extended system $|\Psi_{vh}\rangle$. Then, this approximation can be used to efficiently sample the extended Hilbert space, and estimate the expectation value of any operator of interest by tensoring it with the identity for the hidden units and computing the expectation value via Monte Carlo on the variational state.

D.2 Relation to the cRBM architecture and universality of RBQS

Here, we show that applying a projection operator to the hidden partition collapses the visible spins into a pure state. Then, we demonstrate that specific projections reduce the RBQS and DBQS Ansätze to the cRBM and DBM wavefunctions of Refs. [1, 8], establishing that the DBQS framework generalizes these earlier models.

Given a state of the hidden spins $|\chi_h\rangle$, applying the projection operator $P_\chi = \mathbb{I}_v \otimes |\chi_h\rangle \langle \chi_h|$ to a generic state of the extended system $|\psi_{vh}\rangle$ yields

$$\begin{aligned} (\mathbb{I}_v \otimes |\chi_h\rangle \langle \chi_h|) |\psi_{vh}\rangle &= \sum_{\mathbf{v}, \mathbf{h}} |\mathbf{v}, \mathbf{h}\rangle \langle \mathbf{v}, \mathbf{h}| (\mathbb{I}_v \otimes |\chi_h\rangle \langle \chi_h|) \sum_{\mathbf{v}', \mathbf{h}'} |\mathbf{v}', \mathbf{h}'\rangle \langle \mathbf{v}', \mathbf{h}'| \psi_{vh} \\ &= \sum_{\mathbf{v}, \mathbf{h}, \mathbf{v}', \mathbf{h}'} |\mathbf{v}, \mathbf{h}\rangle \langle \mathbf{v}| \mathbf{v}'\rangle \langle \mathbf{h}| \chi_h\rangle \langle \chi_h | \mathbf{h}'\rangle \langle \mathbf{v}', \mathbf{h}' | \psi_{vh} \\ &= \sum_{\mathbf{v}, \mathbf{h}, \mathbf{h}'} |\mathbf{v}, \mathbf{h}\rangle \langle \mathbf{h}| \chi_h\rangle \langle \chi_h | \mathbf{h}'\rangle \langle \mathbf{v}, \mathbf{h}' | \psi_{vh} \\ &= \sum_{\mathbf{v}} \left[\sum_{\mathbf{h}'} \langle \chi_h | \mathbf{h}'\rangle \langle \mathbf{v}, \mathbf{h}' | \psi_{vh} \right] |\mathbf{v}\rangle \otimes \sum_{\mathbf{h}} \langle \mathbf{h} | \chi_h\rangle |\mathbf{h}\rangle \\ &= \sum_{\mathbf{v}} \left[\sum_{\mathbf{h}'} \langle \chi_h | \mathbf{h}'\rangle \langle \mathbf{v}, \mathbf{h}' | \psi_{vh} \right] |\mathbf{v}\rangle \otimes |\chi_h\rangle. \end{aligned} \quad (59)$$

Now consider the special case where $|\psi_{vh}\rangle = |\Psi_{RBQS}\rangle$ is an (unnormalized) RBQS wavefunction as in Eq. (19), and

$$|\chi_h\rangle = |\rightarrow_h\rangle = \sum_{\mathbf{h}} |\mathbf{h}\rangle$$

is the (unnormalized) uniform superposition over all hidden configurations. Then, the projected state becomes

$$\begin{aligned} (\mathbb{I}_v \otimes |\rightarrow_h\rangle \langle \rightarrow_h|) |\Psi_{RBQS}\rangle &= \sum_{\mathbf{v}} \left[\sum_{\mathbf{h}'} \exp \{ \mathbf{v}^T W \mathbf{h}' + \mathbf{b}^T \mathbf{v} + \mathbf{c}^T \mathbf{h}' \} \right] |\mathbf{v}\rangle \otimes |\rightarrow_h\rangle = \\ &= \sum_{\mathbf{v}} \left[\exp \left\{ \sum_{i=0}^{N_v} b_i v_i \right\} \prod_{j=0}^{N_h} \cosh \left\{ c_j + \sum_{i=0}^{N_v} v_i W_{ij} \right\} \right] |\mathbf{v}\rangle \otimes |\rightarrow_h\rangle . \end{aligned} \quad (60)$$

Hence, the state in which the visible spins are found exactly matches the cRBM wavefunction introduced in Ref. [1]. Similarly, projecting a DBQS wavefunction with two hidden layers onto $|\rightarrow_h\rangle$ yields the DBM wavefunction of Ref. [8], where the visible amplitudes result from marginalizing over intermediate hidden layers. This establishes that both the cRBM and DBM can be viewed as specific subclasses of the RBQS and DBQS Ansätze, respectively. Remarkably, since the DBM wavefunction family has been proved to be a universal approximator of quantum states [74], the DBQS inherits this expressive power, while potentially representing complex quantum states more effectively.

Following is the discussion that applying the projection operator to a variational approximation of the ground state of the extended system does not collapse, in principle, the visible spins onto a good approximation of the ground state of the physical system.

Applying the projection to an entangled ground state $|\Psi_{vh}(G)\rangle = \sum_{a,\mathbf{h}} G_{a\mathbf{h}} |\Psi_v^a\rangle \otimes |\mathbf{h}\rangle$ gives

$$\begin{aligned} P_\chi |\Psi_{vh}(G)\rangle &= \sum_{\mathbf{v}} \left[\sum_{\mathbf{h}} \chi_h^*(\mathbf{h}) \langle \mathbf{v} | \sum_a G_{a\mathbf{h}} |\Psi_v^a\rangle \right] |\mathbf{v}\rangle \otimes |\chi_h\rangle \\ &= |\Psi_v(\tilde{G})\rangle \otimes |\chi_h\rangle , \end{aligned} \quad (61)$$

where the new coefficients $\tilde{G}_a = \sum_{\mathbf{h}} \chi_h^*(\mathbf{h}) G_{a\mathbf{h}}$ define a pure state of the visible subsystem (we used $\sum_{\mathbf{v}} |\mathbf{v}\rangle \langle \mathbf{v}| = \mathbb{I}_v$). Thus, the projection effectively extracts a pure ground state for the visible spins from the original entangled ground state. In practice, however, the optimized variational model typically converges to a variational approximation of the ground state, such as

$$|\psi_{vh}(G, E)\rangle = \sum_{a,\mathbf{h}} G_{a\mathbf{h}} |\Psi_v^a\rangle \otimes |\mathbf{h}\rangle + \sum_{b,\mathbf{h}} E_{b\mathbf{h}} |\Phi_v^b\rangle \otimes |\mathbf{h}\rangle ,$$

where the excited-state weights are small, i.e., $\sum_{a,\mathbf{h}} |G_{a\mathbf{h}}|^2 \gg \sum_{b,\mathbf{h}} |E_{b\mathbf{h}}|^2$. In this case, projecting the state yields

$$P_\chi |\psi_{vh}(G, E)\rangle = \left[\sum_a \tilde{G}_a |\Psi_v^a\rangle + \sum_b \tilde{E}_b |\Phi_v^b\rangle \right] \otimes |\chi_h\rangle ,$$

with

$$\tilde{G}_a = \sum_{\mathbf{h}} \chi_h^*(\mathbf{h}) G_{a\mathbf{h}}, \quad \tilde{E}_b = \sum_{\mathbf{h}} \chi_h^*(\mathbf{h}) E_{b\mathbf{h}}.$$

Although the unprojected state is dominated by ground-state components, the projection may amplify excited-state contributions if χ_h has large overlap with hidden configurations where $E_{b\mathbf{h}}$ is non-negligible. Therefore, the projected visible state is not guaranteed to remain close to a ground-state of the physical Hamiltonian.

D.3 Dataset-free initialization of the DBQS parameters

Typical uses of DBMs in machine learning rely on greedy layer-wise pretraining [46]. In the NQA setting such pretraining yields no benefit: the first few optimization steps rapidly overwrite any pre-trained weights, effectively undoing the layer-wise initialization long before the algorithm approaches the hard

part of the evolution. To our knowledge, however, dataset-free initialization of DBMs has not been systematically explored. Motivated by Ref. [75], which shows that Xavier initialization [76] improves training of canonical RBMs, we adopt a Xavier-normal initialization scheme for the variational parameters: all biases are set to zero, and the real and imaginary parts of the weights connecting layer (l) and layer ($l + 1$) are sampled independently from a normal distribution with 0 mean and variance $a/(N_{(l)} + N_{(l+1)})$, where $N_{(l)}$ is the number of units in layer l and a is a gain factor (here fixed at $a = 10^{-2}$).

If a BQS parameters are drawn from $\mathcal{N}(0, \sigma^2)$, the ensemble-averaged log-amplitude $\mathbb{E}[\log \psi(\mathbf{x})]$ is exactly zero for any spin configuration $|\mathbf{x}\rangle$, and the variance scales linearly with σ . Hence, for sufficiently small gain factor the untrained network already provides a reasonable representation of the $|\rightarrow\rangle$ state.

D.4 Comparison of block Gibbs chain and MH chain

In Markov Chain Monte Carlo (MCMC) sampling, successive samples are typically correlated due to the Markovian nature of the process. This autocorrelation reduces the efficiency of the sampling procedure, as it decreases the number of effectively independent samples obtained from the chain. The degree of this dependence can be quantified using the integrated autocorrelation time (IAT), which provides a measure of the number of iterations required, on average, to produce an effectively independent sample.

Let $\{\mathbf{x}_t\}_{t=0}^{N-1}$ denote a stationary Markov chain, and consider a real-valued function of the chain state $f(\mathbf{x})$. The autocorrelation function at lag t is defined as

$$\rho_f(t) = \frac{\mathbb{E}[(f(\mathbf{x}_0) - \mu)(f(\mathbf{x}_t) - \mu)]}{\sigma} \quad (62)$$

where μ and σ denote the mean and variance of $f(x)$ on the chain. The IAT can be estimated by

$$\tau_f = 1 + 2 \sum_{t=1}^{+\infty} \rho_f(t) \quad (63)$$

In practice, the infinite sum is truncated at a lag where $\rho_f(t)$ becomes negligible. When considering multiple functions, one should consider the highest IAT across all functions to ensure reliable statistical estimates. In the context of spin systems, where the state is represented by spin variables, the function of interest are typically the individual spins: $f_i(\mathbf{x}) = x_i$. For MCMC algorithms employing local update rules, such as the single-spin flip Metropolis-Hasting chain, the autocorrelation time often scales at least linearly with the system size. Specifically, if only a fraction α of the N_s spins are updated in each iteration, at least N_s/α updates are required to provide each spin an opportunity to change, thereby influencing the overall autocorrelation time. [Moreover, Metropolis-Hasting typically has an acceptance rate about 25% further increasing the autocorrelation times, as opposed to the Gibbs chain that accepts every update automatically.](#)

We run both Gibbs chains and MH chains for $N = 10$ realizations of DBQS with different system sizes, 3 layers and hidden units density per layer of 1. Figure 6 shows the average IAT for untrained DBQS with 3 hidden layers and hidden units density per layer of 1 for different system sizes. As expected, we find that the block Gibbs chain systematically converge faster and have autocorrelation times of order $O(1)$ compared to the single spin flip metropolis hasting chain that has IAT of order $O(N)$.

E Algorithmic details on Neural Quantum Annealing

This section provides supplementary technical details on NQA. In particular, we specify the exact implementation of our code and list all tunable hyperparameters, commenting on their effect on algorithmic behavior and computational cost. From a broad algorithmic perspective, NQA is an advanced Monte Carlo method: its runtime scales polynomially, and it does not guarantee finding the exact solution in a strict mathematical sense. Despite the lack of performance guarantees, for many spin glass problems described in Section 8, it successfully finds the best or best-known solution, or a good approximation. As for any Monte Carlo algorithm, the quality of this approximation can be systematically improved through careful hyperparameter tuning.

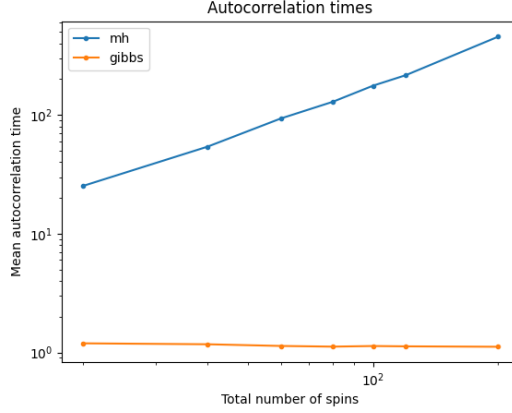


Figure 6: Comparison of the magnetization IAT between the single spin flip Metropolis-Hasting chain and the block Gibbs chain.

E.1 Algorithmic overview

As discussed in Section 7, NQA combines SR with momentum acceleration and persistent Gibbs chains enabled by our novel DBQS architecture. The update step is shown in Algorithm 3, and recursively executed throughout the optimization. It comprises three actions:

1. **Sampler update:** Advance the persistent chains with a few block-Gibbs sweeps and draw a set of configurations;
2. **Natural-gradient evaluation:** Compute the natural gradient;
3. **Parameter update:** Update the variational parameters and the optimizer state.

This update step is embedded into the full NQA optimization showcased in Algorithm 4, which comprises the following actions:

1. **Initialization:** The parameters of the DBQS are drawn from the Xavier-normal distribution as described in Section D.3. An ensemble of block-Gibbs chains is then initialized and thermalized.
2. **Pre-training:** The Ansatz is first trained to approximate the ground state of the initial Hamiltonian at $s_0 = 0$. When the initial Hamiltonian is chosen as $H_0 = -\sum_i \sigma_i^x$, this step can be skipped because, as argued in Section D.3, the Xavier initialization already yields an accurate approximation to its ground state.
3. **Iterative annealing:** For each intermediate annealing point s_j , we perform a few NQA update steps. In practice, a single step for each value of s_j ($j = 1, \dots, N_A - 1$) is usually sufficient to keep the state close to the variational adiabatic path and successfully approximate the final target state.
4. **Fine-tuning:** Finally, additional update steps are carried out at $s_{N_A} = 1$ in order to suppress any residual excited-state components of the wavefunction.

Algorithm 3 NQA_UpdateStep

Require: Variational parameters θ ; Persistent Markov chains $\mathcal{C}$; Optimizer state α .

Ensure: Updated parameters θ ; Updated Markov chains $\mathcal{C}$; Updated optimizer state α .

- 1: Update the Markov chains $\mathcal{C}$ and sample a set of configurations
 - 2: Compute the natural gradient $\mathbf{g} \leftarrow \mathbf{g}_\theta E(\theta)$
 - 3: Compute the updates for the parameters and the new optimizer state $(\delta\theta, \alpha) \leftarrow \text{opt}(\mathbf{g}, \alpha)$
 - 4: Update the variational parameters $\theta \leftarrow \theta + \delta\theta$
 - 5: **return** $\theta, \mathcal{C}, \alpha$
-

Algorithm 4 NQA

Require: Number of warm-up steps N_W ; Number of annealing steps N_A ; Number of updates per annealing step N_U ; Number of fine-tuning steps N_F ; Optimizer $\text{opt}(\mathbf{g}, \alpha)$ (SGD with Momentum); Variational Ansatz $\psi(\mathbf{x}; \boldsymbol{\theta})$ (DBQS).

Ensure: Optimized parameters $\boldsymbol{\theta}$

```
1: Initialize parameters  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$  ▷ Initialization
2: Initialize auxiliary state of the optimizer  $\alpha \leftarrow \alpha_0$ 
3: Initialize and thermalize Markov Chains  $\mathcal{C}$ 
4:  $H(s) \leftarrow H_0$ 
5: for  $n = 0$  to  $N_W - 1$  do ▷ Pretraining
6:    $\boldsymbol{\theta}, \mathcal{C}, \alpha \leftarrow \text{NQA\_UpdateStep}(\boldsymbol{\theta}, \mathcal{C}, \alpha)$ 
7: end for
8: for  $j = 1$  to  $N_A - 1$  do ▷ Iterative annealing
9:    $s \leftarrow j/N_A$ 
10:   $H(s) \leftarrow sH_T + (1-s)AH_0 + s(1-s)BH_C$ 
11:  for  $n = 0$  to  $N_U - 1$  do
12:     $\boldsymbol{\theta}, \mathcal{C}, \alpha \leftarrow \text{NQA\_UpdateStep}(\boldsymbol{\theta}, \mathcal{C}, \alpha)$ 
13:  end for
14: end for
15:  $H(s) \leftarrow H_T$ 
16: for  $n = 0$  to  $N_F - 1$  do ▷ Finetuning
17:   $\boldsymbol{\theta}, \mathcal{C}, \alpha \leftarrow \text{NQA\_UpdateStep}(\boldsymbol{\theta}, \mathcal{C}, \alpha)$ 
18: end for
19: return  $\boldsymbol{\theta}$ 
```

E.2 NQA Hyperparameters

In the following, we enumerate the tunable hyperparameters of NQA, organizing them into five groups based on their role in the algorithm.

Annealing Hyperparameters These parameters characterize the annealing schedule and the parametric Hamiltonian $H(s)$ in Eq. (30).

- Number of annealing steps N_A : it fixes the number of discretized values of the annealing parameter along the schedule, precisely given by $0 = s_0 < s_1 < \dots < s_{N_A} = 1$.
- Number of warm-up update steps N_W : number of `NQA_UpdateSteps` to perform for $s_0 = 0$ to prepare the initial ground-state.
- Number of updates steps per annealing step N_U : number of `NQA_UpdateSteps` performed for each intermediate value of s_j , for $j = 1, \dots, N_A - 1$. Using a number of updates per step equal to one is sufficient for most classical and quantum applications, by selecting a sufficiently large number of annealing steps N_A , as anticipated in Section 7.
- Number of fine-tuning update steps N_F : number of `NQA_UpdateSteps` performed at $s_{N_A} = 1$. Whereas this final refinement does not seem to yield any performance improvement on classical problems, it empirically improves the performance for quantum ground states.
- Annealing field scale $A > 0$: multiplicative prefactor of the initial Hamiltonian H_0 . Typically, during the annealing dynamics, the system undergoes a phase transition (a crossover, for finite systems) from a paramagnetic to a spin glass phase (additional phase transitions, such as perturbative crossings, may also occur). Heuristically, NQA is expected to work better when such a transition occurs somewhat close to $s \approx 1/2$. Indeed, if the transition happens for $s \approx 0$, there is almost no difference between NQA and standard NQS optimization on the target problem Hamiltonian, since the hard regime is immediately reached. On the contrary, when the transition happens for $s \approx 1$, the variational Ansatz may not have enough time to relax on the target ground-state in the hard regime, and computational resources may be wasted in the easy

regime. Correctly tuning A such that the transition occurs neither too soon nor too late during the annealing can be achieved automatically through hyperparameter optimization.

- Catalyst field scale B : multiplicative prefactor of H_C , included for the same reason as the annealing field scale. If we set $B = 0$, no catalyst is utilized.

Optimizer Hyperparameters As stated in the main text, we use natural gradients with momentum. Hence, the optimizer hyperparameters are the learning rate $\eta > 0$, momentum $0 \leq \mu < 1$, and diagonal shift λ_d . The diagonal shift does not seem critical for the performance of the algorithm. On the contrary, selecting optimal learning rate and momentum values is crucial and should be done automatically via hyperparameter optimization, as the optimal values strongly depend on the target problem and on the NQS architecture.

DBQS Hyperparameters

- Number of hidden layers N_L : number of hidden layers in the DBQS architecture defined in Section 6.
- Units density per layer α : ratio between the number of hidden spins in each hidden layer (assumed to be layer-independent) and the number of visible (physical) spins in the visible layer.
- Parameters type (real or complex): whether to use real or complex variational parameters θ . Real parameters can be used for stoquastic Hamiltonians as the instantaneous ground state can be represented by real and positive amplitudes. Using real parameters enforces this inductive bias, reduces the capacity of the Ansatz, and may yield reduced computational cost and runtime.

Block Gibbs sampler Hyperparameters

- Number of samples: number of samples to generate to compute observables, such as the local energy.
- Number of chains: number of persistent block Gibbs chains to run in parallel. We recommend having one chain per sample, if possible, as this yields a faster runtime and guarantees that the samples are uncorrelated.
- Number of thermalization steps: number of Gibbs chain steps to thermalize the chains at the beginning of the algorithm. (Recommended values may be 100 – 1000).
- Number of sweeps: number of Gibbs chain steps to perform in between drawing successive samples on the chains. (Recommended 10 – 100).

E.3 Hyperparameters tuning

Finding the optimal hyperparameters for the algorithms is challenging as they correlate in a non trivial manner. The optimal choice depends from the problem at hand, and as for most of machine learning algorithm there is no universal set of hyperparameters that works every time for every problem. To find the optimal hyperparameters for a given problem we use Optuna [62], a framework for black box optimization commonly used in the machine learning community for hyperparameters tuning. The specific algorithms we use are the Tree-structured Parzen Estimator (TPE) [77] and the Covariance Matrix Adaptation Evolution Strategy (CmaEs) [78] depending on the computational budget and the parallelization capabilities available.

TPE in short: on each trial and for each parameter fits two Gaussian Mixture Models $l(x)$ and $g(x)$ the first one to the set of values associated to the best objective value and the second to the remaining values. Then, it chooses the value that maximizes $l(x)/g(x)$.

CmaEs in short: it is an evolutionary algorithm that operates by evolving a population of candidate solutions. It samples a batch of hyperparameter sets from a multivariate normal distribution, evaluates each and then updates the distribution’s parameters based on the top-performing samples. However, it typically require a large number of evaluations to reach its potential.

Typical/Recommended ranges for HPO In our HPO for the classical spin glasses we tune the hyperparameters in the following ranges (catalyzed protocol).

- Number of annealing steps: ("int", 1e3, 1e6, "log"),
- Annealing field scale: ("float", 1e-1, 1e1, "log"),
- Catalyst field scale: ("float", 1e-1, 1e1, "log") (or set 0 if real parameters),
- Number of hidden layers: ("int", 1, 3, "linear"),
- Unit density per layer: ("float", 0.25, 4, "log"),
- Num samples: ("int", 2**3, 2**7, "log"),
- Num sweep steps: ("int", 2**2, 2**7, "log"),
- Learning rate: ("float", 1e-4, 1e0, "log"),
- Momentum: ("float", 0.0, 0.9, "linear"),
- Diagonal shift: ("float", 1e-9, 1e0, "log"),

While keeping fixed

- Number of warm-up update steps: 1
- Number of update steps per annealing steps: 1
- Number of finetuning steps: 1
- Number of chains equal to the number of samples
- Number of thermalization steps

For the TFSK we fix most of the parameters and optimize only

- Unit density per layer: ("float", 0.25, 4, "log"),
- Number of annealing steps: ("int", 1e3, 1e6, "log"),
- Num sweep steps: ("int", 1e3, 1e5, "log"),
- Learning rate: ("float", 1e-4, 1e0, "log"),
- Momentum: ("float", 0.0, 0.9, "linear"),

while keeping the others fixed to

- Number of warm-up update steps: 10
- Number of update steps per annealing steps: 1
- Number of finetuning steps: 100
- Number of chains equal to the number of samples
- Number of thermalization steps 2**7
- Annealing field scale: 1
- diagonal shift 1e-3
- num samples 201

E.4 Computational complexity

The computational complexity of NQA is linear in the number of calls of the `NQA_UpdateStep` function, which amounts to $N_W + N_F + N_A N_U$. The first two terms in the sum are usually negligible, as is the computational cost of the rest of the code; hence, the overall computational cost is essentially proportional to $N_A N_U$ times the computational cost of a `NQA_UpdateStep` function call. The latter comprises three contributions; we will now analyze them one by one.

First, propagating each persistent block Gibbs chain for N_{sweeps} sweeps through a DBQS with N_L hidden layers requires $O(N_{\text{sweeps}} N_L)$ matrix multiplications of matrices of size $O(N_v)$ times a vector of size $O(N_v)$. An operation that has computational complexity of $O(N_v^2)$, for a total complexity of $O(N_{\text{sweeps}} N_L N_v^2)$, and updating all the chains costs $O(N_{\text{samples}} N_{\text{sweeps}} N_L N_v^2)$.

Second, the complexity of computing the natural gradients via SR scales as $O(N_{\text{params}}^3)$ or via minSR $O(N_{\text{params}} N_{\text{samples}}^2 + N_{\text{samples}}^3)$ [2] for the matrix pseudoinverse. Since a DBQS has number of parameters $O(N_{\text{spins}}^2 N_{\text{layers}})$, the SR route costs $O(N_{\text{spins}}^6 N_{\text{layers}}^3)$, and minSR reduces this to $O(N_{\text{spins}}^2 N_{\text{layers}} N_{\text{samples}}^2 + N_{\text{samples}}^3)$. There is also the calculation of the logarithmic derivatives that is just computing averages from the samples previously generated $O(N_{\text{params}} N_{\text{samples}})$. And the local energies, that as mentioned in the main text they involve a matrix multiplication $O(N_v^2)$ and a scalar product $O(N_v)$. Total cost via minSR is $O(N_{\text{params}} N_{\text{samples}}^2 + N_{\text{samples}}^3)$.

Third, evaluating the local energy and related observables is $O(N_{\text{spins}}^2)$, while the subsequent momentum-SGD update of the parameters is negligible.

Collecting all terms, the per-step complexity for the practical (minSR) implementation is

$$O(N_{\text{spins}}^2 N_{\text{layers}} (N_{\text{samples}}^2 + N_{\text{sweeps}}) + N_{\text{samples}}^3), \quad (64)$$

which remains tractable for the modest $N_{\text{samples}}, N_{\text{sweeps}} \ll N_{\text{spins}}^2$ typically required in practice.

E.5 Visualization of the NQA trajectory

To visualize the NQA trajectory we use PCA to reduce the dimensionality. After a simulation, we have the parameters at each step θ_s . PCA of θ_s (treating the as real parameters), yields the directions in parameter space that explain most of the variance. We project the trajectory onto this plane, and plot the loss landscape with the corresponding reconstructed parameters.

F maybe to do

histo of the number of steps needed to find the best config??

The visualization technique used to make Fig. 1 could actually used for something more than just making a figure: it allows to study how the energy landscape evolves locally, potentially useful for systematically studying different annealing schedules/hamiltonian. We can already see how the the energy landscape flattens out around the "transition point", and this is probably where it fails, and all the discussion about the phase transition and its problems that I am sure you are familiar about. Now we have the possibility to not only visualize, but mathematically quantify how different choices of annealing Hamiltonians influence the procedure, potentially shining new insight even for real physical implementations. One thing that we tried in the past was to see if using the parametric Hamiltonian

$$H = \cos(s)H_A + \sin(s)H_t, \quad s \in [0; 2\pi],$$

would yield a better protocol (as in more stable/faster numerics), based only on the intuition of trying to keep the instantaneous energy more constant. Now we can better study how it influence the protocol.

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